

Untitled

Author: Hemat Wander

Date: 16h

@Aarjav Jain @Krish D

Going through checklist to finalize PCB: Before ordering, the last things I want to do are check over all of the notes I've written to ensure nothing was missed.

Some changes:

- I changed the Vin capacitance to a larger value to better support switching during scrutineering. if this later isn't desired, we should be able to revert to a 100nF as before.

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- I removed the polygon pours for the top and bottom signal layers from last time. This is because we want to keep the HV parts of the PCB (on the top and bottom layers) as isolated as possible.

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- I made all the vias tented, to prepare for ordering.

- Common mode capacitance, I previously removed the DNP common mode capacitors due to them adding a stub for possibly no benefit. As Micha mentions, the presence of the common mode choke makes it kind of redundant, however it might be helpful later down the line if we are experiencing noise issues we can't help. Plus we can also try cutting the trace later down the line if needed. [Related Masterboard Update](#)

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I thus added in the capacitors.

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- Added this silkscreen

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Going through the previous reviews

I addressed everything in this [review](#), remaining points:

I also addressed all of the points / made comments for all the other reviews.

PCB checklist:

I went through this [PCB checklist](#), which made me re-evaluate the silkscreens, and vias sizing.

Other points to review:

- Currently I have the input for the buck converter connected to the module 16 after it has been connected by the autoconnection circuitry. Looking at this again, I'm considering if I need to do this, or if I should connect it to before the autoconnection circuitry? As the PFETs should only have milliohms of resistance, and we have a 100 ohm resistance in the buck converter input path anyways, it shouldn't matter. Also, I'm not sure how VDRIVE will behave when the ADBMS1818 is not powered. Thus, its best to leave it as after the PFET, thus everything gets powered at the same time.

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- I increased text height to 1.5mm

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I checked that the IC's all have the correct pinnout - ADBMS1818 - SM91501ALE - SN74LV4052AQDYRQ1 - LCC110STR - SN74LVC1G3157DCKR - PFETs and NFETS

Added bypass pathways for VREG and VREF, so they don't get cut off at the edge of the board and cause overheating.

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- Added a ground path under the returning voltage lines here at the cost of being closer to isoSPI.

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- I realize I accidentally reversed the channel A and B of the isoSPI, so I needed to reverse them and change the silkscreen accordingly.

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- The old altium has a bunch of stitching vias for VREG, not sure why they added this, or if this is something for me to consider. Currently, I have all my power routed through traces.

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Remaining Points (these are not finished in this update): Bigger changes: - Should we change how the 16 voltage taps are connected to the ADBMS1818

- Should we connect Vin of the ADBMS1818 to the internal 16th module always (even when scrutineering)

Smaller considerations:

Discharge timer for the ADBMS1818 (DTEN) -> 0 ohm resistor?

Should we have fusing on VREG

Do we need to fuse the main GND of the slaveboards

Minor points:

- Do we need to make the pinnout match that of the module board

Do we have enough test points?

Are vias sized and multiplied appropriately for the current they must handle?

Is this an issue for HV integrity?

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Untitled

Author: Hemat Wander

Date: 2d

@Aarjav Jain @Krish D Filling the top and bottom layers with a polygon pour:

As this slave board iteration is hopefully wrapping up, one last thing I wanted to add was to fill the top and bottom layers with a polygon pour, to decrease cost (less area needs to be cut away), and more importantly to create a neighboring return path, which will increase the speed and accuracy of the readings, by treating these ADC lines more like a high speed signal. It should reduce the impedance of the lines which should improve reading accuracy.

I wasn't able to find any resources saying if this is a good idea or not, but based on wanting to reduce inductance, this seems like a good idea? Resources: <https://www.analog.com/en/resources/app-notes/an-1142.html>

<https://resources.altium.com/p/how-properly-ground-adcs>

**What will it look like? **

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Choosing a net: There are a few things I could make these polygon pours, either the filtered voltage tape (C1 .. C18), or the input voltage (V1), or the autoconnection voltage (VIN_TOG_1), etc. Furthermore, I could either make the polygon pours be the return path, or the actual path itself.

For the first point, I think making it be the fused voltage makes the most sense (V1) as that will be the raw return path going up to the actual voltage of the cell (similar to a raw GND). Next, I think making it be the return path makes more sense to reduce inductance, as we otherwise wouldn't have a neighboring return path.

I repeated a similar thing on the bottom layer.

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Important note: I am relying on this [HV routing calculator](#), which says that internal conductors (I'm interpreting to mean traces inside the PCB), only need a distance of 0.1mm, and all of our rule distances are at least >0.2mm or >0.254 mm. This is why I think that including this polygon pours for this HV board is okay.

Other Polygons: There was some space left over after pouring the return paths for each ADC layer. Thus, I decided to add a polygon pour for the VREFs. I decided to note add one for VREG, as I thought it might radiate unnecessarily radiate switching noise to the ADC lines. However, VREF2 should be safe to add a polygon pour for.

Other Notes: I rerouted isoSPI again, this time avoiding cross over any of the isoSPI lines, while keeping a <1ps time difference between the adjacent lines.

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Important: I made all stitching vias untented for the sake of reviewing, we need to revert the vias back to being tented afterwards. CC: @Aarjav Jain

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Aarjav Jain (2d)

@Hemat Wander : Sorry, the vias can stay tented I was just wondering if that was the reason I could not see them.

Otherwise looks good!

Untitled

Author: Hemat Wander

Date: 3d

Checking things with the Slave board There are a few important things I need to check with the slave board before ordering.

Organizing cell inputs with 16 inputs going into the 18 input ADBMS1818. Currently I have it so that we are skipping the input C_12, as is shown in the last pages of the [ADBMS1818 datasheet](#) (pg. 87). However, I wrote down in my notes that skipping inputs like this actually makes open wire checking for those lines worse. However, I can't remember where I read it, and I unfortunately did not write down where I read it (should do this next time).

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After scouring through the datasheet, I can't find any information about this, so I will leave it as is. Either way, the most I think we would lose is open wire checking on just one cell input, which should be fine. I think it's better to leave it as what the part of the datasheet I could actually find says.

LED input current I realized that the warning LEDs we are using for idiot proofing only have a valid current range of 20mA - 30mA. Meaning that anything outside this will blow up the LED. I tried looking for LEDs that accept a wider range of currents, but I couldn't really find anything greater than a 20mA range. To be clear, we are not sure exactly what current the LED will experience, as that is dependent on how poorly we plug in the connectors.

In the future, we could try consider using [this LED](#) which I think has the same footprint. It takes 50mA - 70mA of input current. I think it would still produce some light below 50mA.

CC: @Krish D @Aarjav Jain

Aarjav Jain (3d)

@Hemat Wander

Hemat Wander (2d)

- Again, I couldn't find where it said it was an issue. However, I think it is an issue because the open wire checks uses 2 neighboring voltage taps (say C11 and C12) to push and pull current respectively, to determine if one of the lines is not connected.

- If we increase the series resistance, then it will be harder (or maybe impossible) to light up the LED with a lower reverse voltage being applied. Depending on how incorrectly we connect the voltage connectors, we could have a reverse voltage ranging from 8V to 100V+ (etc.) Reasonably, the largest voltage drop we would see is like ~35V though.

Aarjav Jain (2d)

@Hemat Wander: I see. Consider harness length as a factor for how badly we could actually plug in connectors. Use that to guide the resistance value. Then avoid the overcurrent situation.

Untitled

Author: Hemat Wander

Date: 3d

Continuing Checking Slave board + Other Related Tasks:

Kind of interesting change:

It turns out that this circuitry for reverse voltage protecting the LEDs doesn't work. Normally, these diodes would be supplied with 4.2V in reverse, meaning that the LED would be taking 2.6V-4.2V of reverse voltage.

The LED datasheet says that it normally can withstand 12V of reverse voltage, meaning that we would normally be safe, however, if we plug in the wrong connector such that we exceed 12V, the LED will break (In some way, I'm not sure exactly what).

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As a *side note*: The reason I thought this circuitry would work before, was that the diode before the LED would block the voltage from passing. However, as a general principle, the order of series components doesn't matter, as all you care about is that the current going through them is the same. In this case, both the diode and LED and diode would experience some split of the voltage depending on their exact bias curves.

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As this circuitry is non critical, we can either keep it as it is, or if we want to avoid the chance of the LED breaking, I'm not entirely sure what voltage at which this would occur. **To be clear, the problem statement here is non-critical, meaning we can just leave the circuitry as is. The diode would protect the cells from anything bad happening nominally. I also acknowledge that procedural changes to stop plugging in connectors incorrectly is another solution. **

To fix this issue, we brainstormed a couple of options, most of which wouldn't work. We got all of BMS + some of BTM to help brainstorm this relatively harmless issue(fun stuff):

- One solution @Deev Shah suggested was using different connectors to avoid needing to idiot proof all together, but I want to avoid using a variety of connectors for ease of harnessing, + if we have space for these components this circuitry is a somewhat elegant solution in my mind.
- A common solution you will see online is this one, used to bypass the LED in the reverse direction. The only issue with this is that the high quiescent current draw, as this will always be active.

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- I tried to create this circuit, which uses the 6.8V Zener's we already had, to make it so there is no quiescent current draw during normal operation, however this will likely bypass the LED with a lower voltage drop, meaning that the LED might not have enough of a forward voltage drop to turn on. ([Zener](#) forward voltage is much lower than LED's voltage). This might still be an option, but its riskier.

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- Another idea was something like the following, where we use a PFET to protect against the wrong direction voltage. Using a PFET in this case doesn't work when $V_{15} - V_{14} > 20V$, as we will exceed the VGS of the PFETs we use (PJAs). *Note the PFET technically is in the wrong place here, but the ideas still applies. *

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- Then, I found this solution, which I was very convinced would work. This solution uses an NFET instead of a PFET to get rid of the large drain source voltage issue. I thought we just need an NFET to withstand 100V drain-to-source. [This was the cheapest one I found on digikey.](#)

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The main issue with this circuit is that it doesn't work. Once again, its important to note that NFETs do not act like perfect switches, and in their off state, they would basically once again act like the diode from the circuit we had originally (meaning the reverse current would travel through the PFET and cause a reverse voltage across the LED). I confirm this with the LTspice simulation below, where I got a higher reverse voltage drop over the LED than the NFET.

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- @Christopher Kalitin Came up with the following solution, where the LED is protected from reverse voltage using a 100 ohm bypass resistor. I think this would work with any arbitrarily large sized resistor up to a reasonable extent (ex. 100k).

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- This is similar to Chris K's idea but uses a diode in reverse instead to make it so the voltage drop over the LED is capped with a logarithmic current-voltage relation as opposed to linear.

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I'm most confident in this solution, and thus I will use it, despite it adding 8 components per board. I think this is worth it.

*Side Note: *

Analog device article related to [reverse voltage protection on modules](#)

*Side Note #2: *

It's obvious we've gotten to the point where there are a lot a lot of components on the slave boards. Do we actually need most of these components, probably not. For example, we could just not have this circuitry, the ESD protection, the buck 0.1 ohm resistor, autoconnection circuitry etc. I think having all of these things though, at the cost of more components and more boards space is worth it given us wanting the slave boards to be very robust and preventative against failure modes.

I acknowledge the risks this adds in terms of more components means more possible failure points, however I think thorough testing will eliminate most of the problems we could see in that regard. However, I agree there is a point where you are going overboard, so let me know your thoughts. @Aarjav Jain @Krish D ?

- Originally the stitching vias Altium's tool was creating were untented, so I had to add this custom query type rule to make them tented.

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Buck Converter:

- For future reference, the buck I chose was purely for the sake of using something we know already works. However there are other options available: [Adjustable output bucks with \$V_{in\(max\)} > 70V\$](#)

[Bucks with 5V output and \$V_{in\(max\)} > 70V\$](#)

- I recalculated the buck convert values, and tried to show my work better. @Aarjav Jain We could go with a 47uF input capacitance (meaning we would need an alum electrolytic capacitor) however this would delay ordering due to the size requirement of an alum capacitor, so in our case I think its battery to go with a smaller ceramic capacitor, which we know from the past should work. However, I do acknowledge this as an oversight and a mistake to not include this previously. (*side note: I'm not sure if it would have fit vertically speaking*)

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- Added pull down resistors for TSEL lines

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- I reverted the isoSPI back to what it was to optimize for the length of the paths being the same (< 1ps difference) at the cost of the lines crossing over one another. Thoughts @Aarjav Jain @Krish D

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BOM:

I created the [BOM](#).

I used this [TVS diode](#) for the buck output, as the one listed there is out of stock now

I calculated around a raw \$267 total, \$134 per board

There is \$40 of fusing I think I can reduce. Along with some other things.

I'm still missing adding the male connectors.

Other notes

- It turns out the 6.8V Zener diodes we were [previously using](#) are running out of stock, so I decided to switch them to the [BZT52C6V8-7-F](#). This required importing and creating the layout for a slightly longer footprint.

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- Changed resistor divider values from 100k to match that from [previous simulation work](#).

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- Changed Diode Pad Width

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- I also tried adding text in the middle of the connector names, but I think its getting messy if I do that, so I won't

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CC: @Krish D @Aarjav Jain

Aarjav Jain (3d)

@Hemat Wander:

Hemat Wander (2d)

@Aarjav Jain

I have now made the stitching vias untended, did you mean all the vias?

I have made a note for the capacitor

For the isoSPi, good point. I assumed that having the isoSPI lines not cross over each other meant that it would be **impossible** to match them to the same time difference. I now have this new routing.

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Sounds good will update today!

Aarjav Jain (2d)

@Hemat Wander

Untitled

Author: Hemat Wander

Date: 7d

Continuing layout + Routing based on DR2 feedback

As the last [chunk of work](#) I did for this was kind of a blur. I want to re-evaluate some of the layout and positioning.

Buck converter: The requirements for this layout was that the components circled below (2 capacitors and the rectifier diode) had to be placed as close as possible to the buck IC. Also, the inductor has to be far enough from the center of the board such that it doesn't interfere mechanically with slaveboard #2's isoSPI harness. With this in mind, the layout below makes sense, and we are able to move it down a little if needed

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I cleaned it up a bit to fit into the box as shown below:

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Scrutineering Circuitry: For these components, there is no mechanical constraint as the solid state relays, connector and other scrutineering specific components will not be needed on slave board #1. Thus, the goal here to place the components in a sensical way to reduce routing complexity and length for the sake of shortening lines to reduce noise coupling. Another consideration is to have the scrutineering test points relatively far from the T_read lines and the buck converter lines to prevent accidental shorting during testing.

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I changed the scrutineering connector to be a mirrored version of the B+ footprint, that way we connect a harness easily.

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I created the following layout, allowing us to easily determine the scrutineering components required for board #2 and the connections we can instead use on board #1.

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Schematic addition: I also added these bypass 0 ohm resistors for slaveboard #1, which will not have any of the scrutineering related circuitry (connector, solid state relays, etc.)

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Debug LED: I changed the debug LED to be closer to the middle so it is visible through the clear panel.

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Temperature Circuitry + Protection Diode Routing: The last thing to finalize the component placement of are these components between the connectors. Currently they are placed rather haphazardly.

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Rerouting IsoSPI: I decided to reroute the isoSPI line as originally I had this little cross over section for the common mode. I'm honestly not entirely sure if this is a bad thing in this case, as it is crossing over a line which is indirectly connected to the same line through the termination resistors. However, I found a topology in which we don't need to do this.

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I changed it to the following to optimize for having no lines crossing over one another, at the cost of more bending of the traces and a 4ps time difference in the differential lines positive and negative.

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@Krish D @Aarjav Jain Please let me know your guys thoughts about this.

I'm honestly not sure what to do about this routing exactly, as I've heard that having the return path as closeby as possible is what should be optimized, but I've also heard that having bends is very bad and should be avoided at all

costs, and I'm not sure how lines crossing over another another plays into all of this.

To be clear, I know each of these things is bad, but I don't know what is worse.

Currently I'm looking through these resources to try and see what to do next, but this is not a big deal and can be changed quickly. [General Differential Pair Video](#)

[Differential Crosstalk article](#)

[Coplanar routing differential pairs](#)

[Mode Conversion](#)

Other changes:

I decided to change the PFETs to the PJAs as I've been kind of shifting around for a long time. The reason for this is kind of singular, to improve the chances of success and robustness of the slave boards at the cost of accuracy of readings during balancing. To this end I will NOT be populating the parallel 100k resistors unless necessary. This decision will be justified in the design doc as well.

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To do so I needed to import the PJAs which didn't have a footprint, so I needed to take the 2N7002-TP Altium footprint and then morph it into the PJAs PFET footprint.

I changed the NFETs to the new standard [NTR4003NT1G](#)

Added a pull up resistor footprint for SPI on the SDO line as shown in this diagram of the datasheet.

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- Aarjav noticed that the transformer in the schematic was reversed from the diagram in the datasheet. The choke should be facing towards the transformer. I have now rectified this in the schematic.

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- Removed the DNP capacitors, as they are just creating big stubs in the circuit for no benefit. I realize the other capacitors are also forming similar stubs, but we **know** those are going to be populated for sure. I think its better to just delete these stubs to avoid these lines serving as antennas.

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- Added mating connector 3D models to the Altium connectors

The (

) for the module-board to slave board and

The (

) for the scrutineering connector.

- I added silkscreen labels for idiot proofing isoSPI connectors.

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- made vias have thermal reliefs

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- added tear drops for all nets

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- Added return vias for the autoconnection circuitry nets

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- Added return vias for the voltage measurements

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- both of the above benefit from return vias because the transient of the autoconnection circuitry occurs very fast, and so we need a good return path to have a proper curve. For the voltage measurements, a good return path is good for better ADC measurements, and better shielding from noise.

- Added stitching vias every 5mm to join the GND planes together.

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Reading up on differential pair resources to find the optimal way to route isoSPI (this is not a priority)

Recomputing buck converter values -> this should only affect the component values not the routing / layout.

- I'm considering filling the top and bottom layers with polygons, as they currently only have traces, however I'm weighing the pros and cons of doing so, as it might lead to more noise coupling, but on the other hand could create a closer return path for the analog lines.

let me know your thoughts.

Aarjav Jain (5d)

@Hemat Wande: TL:DR Just small questions. No major concerns. Please read still.

Otherwise looks good!

Hemat Wander (4d)

@Aarjav Jain

1. I was already using the differential routing tool, do you think I should revert the isoSPI routing to what it already was? (I.e the isoSPI crosses over itself)

2. The capacitors might be helpful for filtering, but the cost is that they create a stub antennae, what are your thoughts on this.

Aarjav Jain (3d)

@Hemat Wander: I am skeptical that the antenna like behavior offsets the benefit of the filtering. What does the application notes say about the filtering? I would say we should add the filtering, test by probing iso spi, and we can cut the trace later if we want to test without.

I would use the routing tool and make slight adjustments. Im not sure if Altium can accurately calculate the impact of switching layers and output that as picosecond difference.

Untitled

Author: Hemat Wander

Date: 10d

Continuing Routing the Changes from the DR2: To make room for the LEDs, when I was routing the buck circuitry, I realized it might be possible to squish the SMD components together to make more space.

Before:

After:

[Missing image: ../../images/image_2819154589.png]

Why I would do this: - Makes room for the reverse connection LEDs without having to resort to 0603 components - Might even look slightly cleaner - keeps ADC lines slightly smaller

Any issues: - No longer any space for the resistance values (not required) - Silkscreen is kind of squished (not a big issue)

Note: You might notice that the silkscreen test is to the left of some components and to the right of others. This is purely for density so the text for the components don't overlap with each other.

I decided not to use the above purely **because its not a requirement (MVP) and I need to get the other stuff done.

**

Changing Connector Another decision I decided to make was going down to 6-pos connectors. and this is purely so we have more space to add the extra components. The only downside is that we have no backup if the 0 ohm resistor on the module board doesn't work, however it should work.

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Then, I needed to decide how to place the temperature components + reversing LED. The options were either to (1) place them between connectors (2) have all the connectors together the middle and then place the components off to the left and right side (like it was previously).

Or some combination of both. An example of what this could look like is shown below. (I didn't end up going with this).

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Benefits of (1) is that the routing is in theory simpler as the connector is more directly closer to the components it is being connected to. It also makes it easier to separate the modules wires to possibly make it look cleaner and

The only issue is that the fuse will be a little more difficult to access.

I decided to go with this option for the reasons above.

Aarjav Jain (10d)

@Hemat Wander: What is the contingency plan for "The only downside is that we have no backup if the 0 ohm resistor on the module board doesn't work, however it should work."?

Hemat Wander (7d)

@Aarjav Jain I'm not quite sure what you mean. The purpose of adding the 0 ohm resistor on the module board was so that we only needed to have 3 wires going from each module board to the slave board. Thus, we are just continuing with that, as opposed to extra line for GND I had before.

Does that make sense?

Untitled

Author: Hemat Wander

Date: 12d

@Krish D @Aarjav Jain Routing Changes from the DR2: This update is about the changes I made from the [previous update](#). It's starting to look like we won't be able to include all of the components on the board. The option we have is to make all of the components smaller (such as using 0603 components).

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Routing the buck converter circuitry: To route this section I relied on this snippet from [the datasheet](#)

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We need to: - Connect the shottky, input capacitor, and output capacitor GNDs to the same point - Place the rectifier, BST and VD capacitors very close to the device - Minimize lead length

To achieve this I layed the components out like this, ensure that the points above were met as much as possible while also having no test points or inductor where it would interfere with the isoSPI connector (as shown below).

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Other notes: - I'm also looking into tightening the SMD components on the board as follows, to make space for the components we have added.

Aarjav Jain (12d)

@Hemat Wander: Go down to 0603 where needed. I would start to do this for components we have to buy anyways (Ex: resistors we already have in 0805). What about adding ~5mm to the board length.

Untitled

Author: Hemat Wander

Date: 12d

@Krish D @Aarjav Jain Revising slaveboard from the DR2 meeting: This update discusses some of the changes we decided to make from the [previous DR2 meeting](#).

Is isoSPI high speed? Should we consider adding additional protections / contingencies: I found this [source](#) for isoSPI, and this [source](#), and this [source](#). From these sources, I learned that we should use CAT5 cable, which has a characteristic impedance of 100 ohms, to thus match the termination resistance we choose. Choosing this cabling gives me me more confidence in our isoSPI line. After doing some reasearch, an alternative common connector would be the RJ45's (ethernet cables) as used on this LTC6820 [isoSPI demo board](#). My conclusion is that connector shouldn't really matter as much as the characteristic impedance will be defined by the cable not the connector. Also note, that this demo board has a **very strange** isoSPI transformer schematic, so in future revisions we could try testing out this sort of circuitry if we wanted. I think this is overcomplicating it for now.

The ADBMS1818 datasheet also suggests using CAT5 cable, but doesn't seem to say anything specific about a connector.

I also found this [DC2350B demo board](#) for the LTC6813. They use the [dlw43sh101xk2l](#) common mode chokes and and the [ESMIT-4180](#) transformers. We could consider using these if needed in the future, but I don't think we should follow the actual ADBMS1818 and LTC6813 application notes for now. They also us 100pF capacitors from each line to GND, which we don't do, but we instead have a capacitor connected to the transformer center tap which should

have a similar effect.

Finally, I found this [DC2792B](#) demo board, which includes this [MMBZ10VAL](#), ESD protection diode setup for lines like this. We can consider adding something similar (this specific part is out of stock on Digi key). Interestingly all of these boards seem to have a slit under their transformer, although I'm not sure why they would do this.

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Routing of LTC6813 demo board:

[Missing image: ../../images/image_2811492476.png]

Routing of the LTC6820 demo board:

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Based on these things, I added the option to include the following TVS diode, however we don't need to include these. [MMBZ10VA-TR](#) (which is somewhat similar to the other TVS diode). I have kept my isoSPI routing the same for now.
@Krish D @Michael Lin What do you think?

Temperature Multiplexer Pathway: (@Gurman Khella read this section)

- I replaced the t_{sense} 1k resistors with fuses such that we will have fuses that break rather than simple current limiting. The failure mode for these fuses should be easily detectable.

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Other than that, we discuss in the DR2 if the GND for the multiplexer needs to be fused. From the reasoning explained below, I decided not to do this.

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Lets take a step back for what we are protecting against. The module boards are being Fed 3 lines (except module 1 and 17 which also have an additional GND, we can ignore this GND because if that GND shorts we are screwed anyways). Now there are 3 **black-box** cases of shorts that can happen, either the B+ and GND short, the B+ and T_{sense} short, or the T_{sense} and GND short. The latter case doesn't matter, as it will just set the T_{sense} line voltage to be GND.

If the B+ and GND shorts, we will blow the GND fuse on the slave board. Then, the voltage on the GND of the thermistor on the module boards will be equal to B+'s voltage (65V in the worst case). The thermistor, will thus have 65V on its GND side, meaning the T_{sense} line will form the voltage divider between the now B+(65V) and VREF2(3V). This will only cause 3mA to flow, and for T_{sense} to be ~30V. Although there won't be any overcurrent directly here, this 30V will break the multiplexers max input voltage of VREG (5V), meaning the multiplexer will somehow break. We can't really predict how this would happen, but worst case the multiplexer shorts internally. In this case you would have 60V across the thermistor, which would lead to 6mA flowing. This should again should be non-catastrophic, we just have to make sure to fix any thermistor issues as soon as we see them.

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In the other case, B+ will short to T_sense, meaning T_sense will directly be pulled up to 65V in the worst case. This would break the temperature readings, and also the multiplexer. From there, its not exactly certain what would happen, but again in the worst case the multiplexer will short internally, meaning we will have 65V shorting directly to GND. meaning we again need a fuse on the line. Which we added. From there, we will have 65V being shorted to VREF2 or VREF3, meaning we will have ~10mA traveling from T_sense into these lines. From a quick search it seems fusing on the order of mA would be very expensive, so I will not add these. We will just have to be serious about fixing temperature reading issues when they arise, I [made a doc](#) I will fill out to make sure we do this.

The worst worst worst case is the multiplexer somehow shorts through the ADBMS1818, but we currently have resistors between them + we have all the fusing shown above, so even this case shouldn't be catastrophic.

What else could the spare GPIO #9 be for? Previously, I discussed using the spare GPIO #9 as a debug LED. But I thought it might be worth considering other possible options for its use.

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The other main idea that came up was to somehow use this GPIO as an auxiliary sense line for detecting something we might want to have information about. For example, being able to detect if a fuse breaks. - It doesn't make sense to put it on the GND fused line, as that line is directly connected to the module boards, and is fused for a reason - We could connect it to the IBIAS ICMP line in order to have an analog reading of the voltage across them (that way we always know what the threshold voltage we chose was).

- I could connect it to VREG to see what the exact voltage of VREG is. Although, I don't know if this will work because VREG is the thing powering the ADC reference.

None of these seem particularly promising, so I will leave this, unless you guys think there are any ideas: @Krish D @Aarjav Jain .

Multiplexing the scrutineering Temperature Sensor: In the DR2, we realized that the Tspare line being connected to a separate GPIO, probably wouldn't make sense philosophically for scrutineering, and based on [Dan's last email response, he would probably agree](#). Conclusion from that email is that we are going to be sticking with multiplexing the voltage into C_17 as opposed to onto C_18.

For the temperature line, we want to do something similar by having a multiplexer for the 16th T_sense line, which either connects to the scrutineering module or the internal module. I found [this 2:1 multiplexer](#) on Digi key which seems to work for this application. The only concern is if I should include a decoupling capacitor for VCC, as there is starting to be limited room on the board. I chose to add one.

Wait.... I'm getting flashbacks. I literally had this multiplexer at one point for this exact purpose, but I deleted it for some reason. Likely because I realized we had spare GPIOs I could use instead.

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Anyways, I added this multiplexer, and then fed the output into one of the main 4:1 temperature multiplexers. I found this source that seems to suggest that cascading multiplexers like this is fine to do. [Cascading Analog Multiplexers](#)

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Should we have RC filtering on the output of the temp multiplexers? Currently we have these RC filters on the output of our Tsense lines, and I want to re-evaluate if we should include these. The general reason for including these is so that we have some sort of filtering right before our ADC reading (from the ADBMS1818), such that any noise picked up between the initial filtering from the thermistor and neighboring 1uF capacitor (such as from the multiplexing) is filtered out. This 100 ohm - 10nF combination is something directly mentioned in the ADBMS1818 datasheet.

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After a quick search, [this article](#), seems to show this configuration but with the addition of a buffering op-amp. Thus, I will keep these capacitors. (In the future we can consider adding a buffer for temp sensing, but it is not an MVP right now). The only thing subject to change is the exact values, but those can also be changed post ordering.

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Revisiting output capacitor for the buck: The [MAX5033](#) datasheet suggests using a capacitor at the output of the buck converter, with a ESR between 100 and 250 milliohms to maximize stability while keeping output ripple low. However, in the application notes it then recommends using a tantalum capacitor, with an ESR of 1.1 ohms, which doesn't make sense. Thus, I will stick with the setup I have with a ceramic capacitor

PLUS, I found [this article](#) that seems to also talk about using a series resistor for the output of a linear regulator, and so it seems fine for me to do this. The article also explains characterizing the frequency to determine this value, but I don't think we need to do that (yet).

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Idiot proofing the connectors

Thinking about this, It isn't exactly clear what at what level to perform idiot proofing on. There are two ways of going about this, either through a hard barrier that makes it impossible to connect the wrong connector, or by creating some indication for when the incorrect connector is plugged in. When would we need to do this? Note that we already have some idiot proofing through the combination of two modules into one connector.

I should mention my bias, which is that I don't want to make the connectors different from one another, as that would look uglier, and be more annoying to bring up harnesses for.

Why would we need idiot proofing? Lets say we connect a connector in the wrong place. The most likely culprit I can imagine (depending on how the wiring turns out) is that we swap two neighboring connectors. Lets say we swapped module 9-10 with module 11-12. The temperature sensing is the same so it wouldn't matter. The voltage lines of course though, are different. (Lets assume 4V for each module).

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At the first stop, we would have the autoconnection circuitry. We would have the following voltages in black, and assuming the resistance is high enough to prevent loading, we would continue to have the following autoconnection circuitry voltages. (black is the module voltages, and red is the voltages of the resistor divider circuitry).

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This means that that module 9 would be activated as the PFET is pulled low, leading to 44V being pushed to C9 while C8 is only 32V. This is a ~12V difference which would be very bad for the ADBMS1818 (maybe not quite as bad because of the 6.8V Zener?).

Simulating in LTspice with this situation, we get a similar result.

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At the end of the day, the only thing we are preventing against by idiot proofing is the ADBMS1818, there ideally shouldn't be any catastrophic issues with plugging in the wrong connector (@Krish D what do you think?)

**** Option #1:**** For this option, we would need to make some of the connectors different from one another so that it is impossible to plug the wrong connector into the wrong place. This could vary from having two types of connectors we alternate every connector, to making every connector different. The latter option is so ridiculous, but its a question of, if we have two different types of connectors, why wouldn't we have more? Again, I don't like the idea of having multiple types of connectors, as it makes it more difficult to make harnesses post bring up.

Looking at [other teams](#), it seems that: - UC Berkely uses all the same connectors on their slaveboards - Midnight sun seems to use the same connectors (can't really tell) - Stanford definitely seems to use the same connector

Option #2: In this case, we would want some circuitry to either protect against or warn against accidentally swapping the connectors.

For protecting against, we would need a way to make the PFET only turn on if the voltages are below some max threshold. I'm not knowledgeable enough to figure out how to do this, and it seems like a lot of overhead. MR. GPT says we could have some other bypass switch connected to the PFET gate which pulls it up if above some voltage threshold, but that seems like overcomplicating it unnecessarily. -> Something to consider for future

For warning against, we would need some sort of LED system for showing when the voltage difference between some module is above some threshold. There are two options for indicating this:

(1) [From some searching](#), I found the idea for this voltage indicator circuitry that only requires 3 components.

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In our case, we need to indicate if the voltage is above some threshold (the maximum each module can be is 4.2V, meaning that the difference in voltage would never exceed this). If we swap around a connector by accident, the voltage would be somewhere from 8V to 12V. We could thus use the 6.8V Zener diodes we are already using.

(2). The other option is to rely on the fact that voltage somewhere will get reversed whenever we swap the connectors (in the example above V₁₀ is at a higher voltage than V₁₁). Thus, we can use diodes facing in that direction to indicate if the connectors are reversed. The only issue with this is that LEDs have very poor reverse voltages on average (~5V). This means, we will have to add another diode in series to deal with the reverse voltages. We can use the same diodes we used for the autoconnection circuitry.

I will go with option #2 as it seems more reliable. To implement this, we need to include this circuitry consisting of 3 components between each connector in reverse fashion. I will use the same red LEDs as are on the HVC.

If we need to have MAX 20mA, and the LED has a 2V drop, and the diode will have a ~0.7V drop during at 20mA. The reverse voltage we would see could vary from 9V in the lowest case to like 20V if we skipped two connectors. In the worst case, we would have 17V across the resistor, meaning we need a resistance of ~820 ohms for 20mA in the worst case. This would produce 7mA in the 9V case.

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In conclusion I chose to go with this warning method, as it will tell us to connect it in the correct order before the autoconnection circuitry allows everything to pass.

Next Steps: - Routing all of these changes - Adding the mating connector 3D models to the Altium

Other notes:

- Looking at the DNP capacitors on the other side of the isoSPI transformer, the routing for them seems to become a big stub essentially, so maybe we should not have this DNP component. @Michael Lin Where did you write down the purpose of this component again?

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Aarjav Jain (12d)

@Hemat Wander: Super awesome update! Keep up the great technical detail as always. Your explanations and resource linking is very important to contributing to a team's success!

Untitled

Author: Aarjav Jain - Electrical Director

Date: 13d

@Hemat Wander : In the rare case that the B- connection to the B+ module board 1 or 17 is lost then the GND of the SB is lost.

CC: @Gurman Khella @Krish D

Hemat Wander (12d)

@Aarjav Jain 1. All of the IC's would lose a GND meaning that they would no longer function, the exact failure mode for each IC would vary, but the most important one would of course be for the ADBMS1818, I think we would have to test to determine the effects of this.

1. There are not explicitly, I think it would be difficult to add a protection mechanism, but we could add a warning mechanism of some kind. Once the warning occurred it would kind of be too late though.

Actually, as the autoconnection circuitry stands, no GND means this would be floating, meaning that the voltage taps should theoretically be isolated from the rest of the slave boards, meaning nothing could turn on. This should be fine?

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Aarjav Jain (12d)

@Hemat Wander

"meaning nothing could turn on." -> Could you elaborate a little more? What exactly happens when that connection is floating because the GND is lost? If the SB loses comms to MST while battery is on then we could get a 'SB COMMS LOST' fault which can be helpful.

Untitled

Author: Hemat Wander

Date: 15d

I will make the changes mentioned in this [DR2](#) meeting, by EOD Sunday, March 8th.

Untitled

Author: Hemat Wander

Date: 16d

Reworking Scrutineering Circuitry: As from this [update](#), we decided to change the scrutineering circuitry, to make it more **likely** to be possible to use during scrutineering.

I changed the scrutineering topology from looking like this:

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to looking like this:

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For routing these changes I decided to move these components for the filtering of module 16s voltage to past the solid state relay.

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This should give us room to move around buck circuitry as suggested to make the test points not be so close together. (Safety hazard as they have a >55V difference at max).

CC:

Untitled

Author: Hemat Wander

Date: 17d

Debug LED: I realize I never made an update on this, but essentially we can use the spare GPIO #9 for a debug LED, that the masterboard can flash on and off. Benefit is that we can directly see if isoSPI is working. This slightly differs from the WDT LED which will always be on if some signal (possibly invalid) was sent through the isoSPI line. This directly shows us if the isoSPI is working enough to turn on and off the GPIO. CC: @Michael Lin @Krish D

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Alternatives: We can set the GPIOs to be separate to power each SSD separately. Although I don't know why we would ever need to do this.

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Untitled

Author: Hemat Wander

Date: 17d

Slave board - Module board Connectors: After doing a [quick search](#), it seems that 8-pos R/A connectors are cheaper than 6-pos R/A connectors, meaning I might as well use 8-pos connectors on the slave boards since they are as big as they are anyways. Thus, on the cell board we probably no longer need to use the 0 ohm resistor as we can just connect the ground connection directly to the connector using one of the spare pins. **However**, we can still use the 0 ohm resistor as well, if we like that solution better.

@Gurman Khella @Krish D

[Missing image: ../../images/image_2801515443.png]

Aarjav Jain (17d)

@Hemat Wander: For connectors like this, price and availability is no problem since we can order many many samples. However, it is important that they do not occupy significant space. Looking at your board, it are not taking up extra space in that there are empty areas of your PCB so that is good.

Can you explain the exact circuitry changes on the module boards if we go with 8pos? CC: @Gurman Khella

Additionally, based on your experience should it be 6pos or 8pos (consider how the wiring changes for the MB to SB harness).

Gurman Khella (17d)

We used the 0 ohm resistor in the first place because then we only need 3 wires coming from the moduleboard to slaveboard. Initially we had started with 4 pins: B-, GND, B+, Tsense; then we simplified using 0ohm resistor so we only need the GND pin on the connector and not the B- pin. Since we only need B- for the 1st and 17th modules to get GND. This got rid of additional wires going between boards and is a simple way to have the connection on the moduleboards. Are you suggesting we revert to using the 4th pin on the moduleboard?

I don't see the point of having an extra pin being used and another wire going between boards. We have the option to use the extra pin, but I think there are more failure points in using the extra pin, than the 0 ohm resistor. It is simpler to just have the connection on the moduleboard so all modules have 3 wires going between slaveboard and moduleboard for consistency. I would like to stick with the 0 ohm resistor, but if there is something I am not considering, let me know.

Gurman Khella (17d)

Decision on my part if you decide to use 8pos. We will stick with the 0 Ohm resistors. I will add a trace from the extra pin on the moduleboards to B-, so we have the option to use the wire to make the connection between B- to GND on slaveboard side if needed.

Untitled

Author: Hemat Wander

Date: 17d

Slave board Scrutineering Circuitry: In a discussion with @Aarjav Jain we realized that our current setup of the scrutineering circuitry might not work for scrutineering because it uses a separate ADC input (C18) in the chain compared to what the cells normally read (C1 - C17). If we wanted to change this, we just need to make the solid state relay switch to C_17 instead of C_18, doing so would require making the changes seen in the diagram below. Essentially, getting rid of R4.65 and the capacitor and diode. Then adding a 68 ohm resistor to make C_17 and C_18 always connected to the same voltage. Making this change should likely not be a big change?

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My opinion: Having the circuitry as we have it **right now**, has one big advantage, where the voltage reading for module 32 can normally be performed from C17, which provides a clean connection. If we change the circuitry as proposed, we will now have to read module 32s voltage through a solid state relay. I don't like this idea because:

1) The routing for the 32nd module reading will be longer and possibly pick up more noise 2) I'm unsure about the effects of reading a voltage through a solid state relay, for our normal cell readings. As there might be issues in terms of noise or some constant offset. I couldn't find any resources online about this, and I don't think it will be too big of an issue, given that the [datasheet](#) says there is only pF of capacitance, and since the maximum resistance is less than that of the 100 ohms needed by the filtering.

In hindsight it might have been a good idea to characterize the noise by testing the circuitry.

Next Steps: Currently I have not changed the circuitry on the PCB. In waiting for [DAN to respond](#). It seems like Steve at least agrees that the 33rd module solution is slick (I think slick means good?). I'm not sure what "The only issue can be that the pack response now is truly lost in the test of the BPS".

@Aarjav Jain Followed up in the email chain about the idea to change from switching on the 32nd module line. If we get a response that we should do something like that I will change it to do so. Currently, I'm not convinced that doing so is any philosophically better than switching on the 33rd module line (given that we will have to change firmware either way) and as we are still using the same BMS measurement chain.

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CC: @Aarjav Jain @Krish D

Note: If none of this works we could revert to a scrutineering version where we solder 3 wires to these through hole pins and extend them out to the control board, that way we can choose to enable what scrutineering circuitry we are connected to with a hardware enable switch outside of the pack (the idea behind this bypass is connecting a jumper cap to connect either module 32 or our scrutineering circuitry). This idea can just be extended to go out to the control board, where we can perform the switching. The downside of the method is that we can't emulate the behavior of the second solid state relay for only providing HV when we need it.

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Aarjav Jain (17d)

@Hemat Wander Thanks for the update. Will need to read it again to understand what exactly those 3 wires are doing. Perhaps you can elaborate more on that?

Regarding Steve's message -> I am confused about Slick because he also said "The only issue can be that the pack response now is truly lost in the test of the BPS." which I agree with. Although "pack" response is extreme to say. We will see when they respond.

Krish D (17d)

@Hemat Wander @Aarjav Jain

I think that this scrutineering circuitry should still be feasible, since it is still literally in the same chain. <-
Although, this is just my interpretation, so we should wait until Dan responds.

@Hemat Wander How time consuming would it be to add this circuitry to LT Spice to see what reading a voltage through a SSR would do the output signal?

We aren't expecting the signal to fluctuate much or drawing excessive current during ADC sampling **unless you are testing balancing**, so I can't imagine there would be many effects as a result of the additional resistance from the SSR.

I agree, lets wait until Dan responds, and if we have to change the circuitry, lets start with a simulation.

Regarding the 3 wire soldered connection, I'd be hesitant since this provides an non-isolated portion of the slaveboard directly exposed on the control board (same as you mentioned). If this is sufficiently covered, this may still be a feasible option, but we should talk about the exact procedure in more detail.

Untitled

Author: Hemat Wander

Date: 18d

Cleaning up routing in Altium: Previously, I completed routing everything in Altium, however I found that the when running the DRC there were 500+ errors present.

I'm suspicious that most of the errors don't really matter. The errors seem to be:

- SMD neck down constraints This rule is about the ratio of a routed track and the SMD component it enters. I found this [forum](#) that suggests having these neck down constraints is for the same purpose as having thermal relief spokes, but also specifically. Although I'm not entirely convinced this is necessary, this is a reasonable thing to have, so I will adhere to this design constraint in the cases where I was directly breaking it.

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before:

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after

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*I did not apply this everywhere, only where the 100% neck down rule was strictly being violated. *

- Minimum allowable angular ring. This one I really am not convinced by because this is the generic via we have set in the design rules (0.4mm hole, 0.8mm diameter), which is used in plenty of other boards.

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Again, I found a forum that says the minimum angular ring constraint for JLC is 0.13mm. This would make more sense with the generic via size we use. After some searching around the [JLC capabilities website](#), I found this requirement for the spacing. Given this minimum constraint, us having 0.254 mm in the rules make some sense. however, currently our annular rings are 0.25mm which is still above this minimum requirement. Thus, in the rules I will set it to be 0.24mm, and for vias to be excluded from the check.

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*I checked with some other boards including the DRD, and they all seemed to have this same issue of angular rings smaller than 0.254mm. *So I think this should be fine. **

Minimum solder mask silver clearance:

I found that the footprint for the temp multiplexers and other ICs had some issues with the solder mask bridge (AKA the distance between pads being too big). Essentially I spent a long time trying to figure out why when I placed a component in the slave board project, it had a different looking footprint, compared to if I placed it in the DRD or HVC projects. ** In slaveboard: **

[Missing image: ../../images/image_2801327212.png]

**

In HVC:**

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It turns out that there is a design rule that makes it so the footprints can be different in different projects even if you are using the same component. Thus, I changed the solder mask expansion for my PCB to match the HVCs of 0.075mm. Note that [JLC](#), specifies a minimum solder mask expansion of 0mm. So this is fine.

Despite changing that setting, there were still some components that broke the minimum solder mask sliver rule. To solve this issue I did two things as follows.

1. Set the rule to be at a minimum of 0.2mm instead of 0.254mm. I can do this because [JLC](#) only specifies a minimum 0.1mm solder mask bridge for 1oz PCBs (@Aarjav Jain is this what we use?), and other boards like the HVC had 0 clearance for this rule and were still able to print ICs with pads really close together like their MCUs.

(Side note: Were there any issues with the printing of either of these boards? I'm asking because they seem to be breaking the solder mask sliver rule since the pads are closer than the 0.1mm minimum set by [JLC](#)?. @Christopher Kalitin @Museok Seo). This [PCBway](#) page doesn't seem to say anything about this specific rule.

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2. For the multiplexer IC I decreased the pad widths greatly and then decreases the solder mask expansion further to 0mil for this IC specifically. (I can set it as low as 0 according to JLC + ICs on other boards already do this). This caused the footprint to go from looking like this:

[Missing image: ../../images/image_2801323842.png]

to looking like this:

- Matched lengths (this seems to include nets that I do not care about matching the lengths of for some reason)

Notes on some other stuff I found: Having a >0.45mm hole-hole clearance is not mentioned in the [wiki](#), so I added this to my rules with a 0.5mm hole-hole clearance.

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Other routing changes:

These 8 temperature sensing traces were previously crossing underneath the buck converter circuitry (which is going to be very noisy due to switching), so I decided to move them away from the switching circuitry. However, there would be two ground planes between them, so this isn't likely a huge issue.

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Changing polygon pour positions: I found this [article from AD](#) that gives some information on the need for grounding in ADC lines (not just high speed lines). I don't really have time to go through it in depth right now, but it seems that we need to treat ADC lines the same as high-speed analog lines. Based on this I decided to keep two fully intact GND planes, and then route each of the VREF2 and VREG power nets on the top layer.

CC: @Krish D

Aarjav Jain (17d)

@Hemat Wander: Why were the vias not made with >0.25mm in the first place? Where the DRC constraints you put in from the Wiki different?

Yes we use 1oz.

There have been no problems regarding silkscreen to pad in the previous boards.

[Missing image: ../../images/image_2801541685.png]

You missed a picture.

Why decrease pad width for multiplexor?

Could you add the pad hole to hole spacing to the Wiki as well? Good find!

I agree with the GND planes concept. @Museok Seo: Take a read of this update for regarding DRC and routing ideas and let us know your thoughts as well!

Christopher Kalitin (17d)

@Hemat Wander

Haven't checked the silkscreen yet, will do that today.

Micheal and I imported [these design rules](#) off for JLCPCB a Github repo, you could do the same to ensure nothing is missed.

These design rules will probably also solve your neck down issue. Also, why have larger traces on the inside? Why not make it all the lower width?

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Museok Seo (15d)

In the Wiki, I believe there isn't a specific section where you include the 0.254mm in the rules for DRC. I actually recall when working with the DRD, the vias that were placed were very large so I recall changing the via sizes to match the previous boards. (I didn't run into this issue last year).

[Missing image: ../../images/image_2809757136.png]

(Two VIA sizes)

I just checked the DRD for your Solder Mask Silver rule, I agree that the gap between solder mask openings are smaller than 0.1mm. However, the boards were still able to be printed as one large opening rather than multiple small openings I believe for the MCU pad. (I can check this in more detail on Saturday). I believe if you have a gap larger than 0.1mm, JLC will be able to create this bridge of solder masks which we currently don't have. I'm curious to see how it will look like for your multiplexer IC!

For the Pad Hole-to-Hole Spacing, you're correct with the distance required from holes > 0.45mm, however I believe the clearance constraint covers this as well (Prevents copper being too close to each other).

I'll look more into the GND-ing article that you have mentioned! It seems to have pretty good information that I'll be looking into about proper grounding methods for analog signals.

Out of curiosity, could you explain about the analog signals you have on your board? Currently on the DRD there are 2 analog signals reading the pedal's voltages. I haven't got to testing the signals from the DRD yet, but I would like to hear the signals that you have on your board so I can implement this for the future revision of the DRD.

Untitled

Author: Hemat Wander

Date: Feb 19

Final talks about the auto-connection circuitry (for real this time): In a [previous update](#), we concluded that we are going to be moving forward with pre-charge circuitry **without** gate capacitors. Another implicit decision I made was the remove the diodes in series with resistors, simply because they were taking up extra space and because they weren't strictly required. The purpose of these resistors is to explicitly change the time-constant of each PFET so they close one after another, however this **IS NOT** strictly necessary (previously I thought it was necessary). The question I want to evaluate today before I begin routing again is to finalize if these resistors are required or not.

[Missing image: ../images/image_2772058505.png]

There are actually 2 parts to this: 1. Either having those series diode resistors or not. 2. Changing the values of the resistor ladder to ensure all the PFET gate-source voltages are at a similar level. (If we didn't do this, some PFETs activation could vary in the level of entire volts, which could drastically change how much current can pass through.

I went through and simulated the 4 possible combinations of these two points. These can be seen below. The first graph for each shows the gate-source voltage of all 16 PFETs. The lower this voltage gets to during activation, the more conductive the PFET becomes. The second graph shows the charging up of the filtering capacitors right before the ASIC (as we can see they charge up in order for graphs with a series resistor).

Removing the series resistors + all ladder resistors are 100k ohms:

[Missing image: ../images/image_2772065984.png]

[Missing image: ../images/image_2772075150.png]

Removing series resistors + **varied resistance of ladder (not 100ks): **

[Missing image: ../images/image_2772084862.png]

[Missing image: ../images/image_2772089817.png]

****Having series resistors + and divider resistors are 100k ohms: ****

Removing series resistors + *varied resistance of ladder (not 100ks): *

****Side note:** By "varying" the resistances in the ladder, I mean changing the values from strictly 100k to something else. I got these values by solving a system of equations for the resistances we need to have equal gate source voltages.

In the case of no series resistors I used:

100k, 100k, 100k, 100k, 105k, 105k, 105k, 107k, 107k, 110k, 110k, 113k, 113k, 117k, 117k, 117k

In the case of the series resistors I used: 100k, 100k, 103k, 103k, 105k, 105k, 107k, 107k, 107k, 110k, 110k, 107k, 107k, 103k, 107k, 169k**

Conclusion:**

We will (1) be including series resistors, and (2) having a varied series resistance ladder. This is because...

(1) We see in general that having the series resistors improves the stability of the charge-up with the filtering, making it so that each module charges up one after another rather than all at once. This is a strictly **MMR** feature, and should not affect our ability to meet our requirements that much, however it provides us more confidence that the ASIC will not be stressed by the ASICs possibly varying in how quickly they each turn on. There are some nuances to this, but it essentially gives us more configurability, as we could "easily" replace these with 0 ohm resistors if we want to revert to an earlier design. The only real cost is more space + components to bring up.

(2) If we compare the PFET gate-source voltages with the varied resistance ladder, and with all 100ks, we easily see that having the varied resistance ladder allows for all the PFETs to be more strongly over the gate-source voltage. This is more important if we ever have to revert to a PFET like the [PJA3471 R1 00001](#) with a high gate-source threshold voltage.

Aarjav Jain (Feb 20)

@Hemat Wander: When/why would we ever: "This is more important if we ever have to revert to a PFET like the [PJA3471 R1 00001](#) with a high gate-source threshold voltage."

Untitled

Author: Hemat Wander

Date: Feb 18

Continuing work to complete the Slaveboard V4 Design:

From the past few updates ([#1](#) and [#2](#)), we made some decisions on what circuitry we are going to keep for the auto-connection circuitry, so the plan for today is to implement those changes. Then, there are also a few other changes I need to add.

Auto-Connection Circuitry:

Added auto-connection circuitry alternative components. Also, I made a [subtab in the doc](#) for information on implementing these alternative pathways.

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Fusing on the GND side:

Currently, we have all of our fusing in place on the 16 module connectors and for the VREF input to the temperature circuitry, going to the slave boards. We also have fusing for the GND of the first module connector, and for the GND of the temperature circuitry. During the recent DR meeting, Saman raised a point that the fusing for the GND side seems redundant and could even be bad because. 1. You now have two places where a fuse could blow for a given short, meaning we have to check multiple places. 2. If the GND fuse blows, that would lead to a floating GND, which could be problematic for most ICs as they need their GND reference to be the lowest voltage.

****Fusing GND: **** First lets evaluate the temperature GND fuses, why did I consider having this in the first place? My concern was that, since we are supplying an "absolute GND" for the sake of temperature sensing to each module board, we are creating this HV relation on every module board between the module voltage and GND. Thus, if those two lines shorted then it would lead to a lot of current surging from the module board through the slave board to GND. (possible causes could be the thermistor leads contacting the module voltage, or something small and conductive shorting the exposed parts together, or degradation of the wires insulation causing the conductive parts to rub together under the forces from the car moving around).

TO BE CLEAR: The module fuses would not protect against a short like this, as it would occur at the module boards and then flow through the slave board GND wires.

[Missing image: ../images/image_2766578630.png]

Normally, this net would have less than 3mA going through it, meaning this is only really for protecting against catastrophic shorts far beyond the nominal current draw. Thus, we shouldn't really ever expect this fuse to blow or to need to replace it. However, if the fuse does blow, we should only expect the temperature sensing nets to be pulled up to VREF2 (@Michael Lin We can do something in firmware to account for this edge case), so this shouldn't pose a problem for the MUX IC itself, as the MUX itself isn't connected to that floating GND.

[Missing image: ../images/image_2766687903.png]

Conclusion: We are going to keep this GND fuse, and I will move it to the center for ease of distribution to each of the modules (as seen below).

****What about fusing the mux inputs? ****

Based on the above logic, we are also feeding the T_sense inputs to each module board, which will be at nominal ~1.5V relative to GND, which creates a HV difference in respect to the module voltages (which can be up to ~70V relative to GND). If these two lines did short, it would raise up the MUX inputs to the module voltages, thereby exceeding the IC's limits. The IC then breaking could either lead to it just not working, or it effectively shorting to GND. This logic is sound, however it doesn't seem feasible to just add a fuse at every Mux input to prevent against this.

Considering this dilemma, I found [this page](#), which among other things suggests adding series resistors with the multiplexers to prevent against the worst of over-voltage shorts. This is a good solution I think, and really the only problem is the lack of space on the slave boards to include these series resistors. Also, I will use 1k resistors, as they will only lead to $70V/1000 = 70mA$ of current draw during a short, and also should have pretty good accuracy. I will also place these resistors in the corners of the ADBMS1818, as shown below.

[Missing image: ../images/image_2766843027.png]

Conclusion: Use 1k resistors in series with every temperature sensing input.

****Fusing the Absolute GND of the slave board?**

****This other fuse we had connected to the absolute GND of the slave boards (from the first module board). This fuse was initially kept as a legacy addition from the previous revision cell-boards which had a fuse at both their B+ and B- connections.**

[Missing image: ../images/image_2766851064.png]

However, re-considering this now, I don't really see a case in which we would need this fuse, as we already have GND side fusing for the temperature circuitry, and at every module positive. Thus, we can safely remove this fuse.

Fusing VREF2: Similarly, I also deleted this fuse because it was no longer being used for anything. (Originally VREF2 was also going to have a direct connection to the module boards).

[Missing image: ../../images/image_2766881314.png]

@Krish D Can you please check that all of these make sense for the fusing?

Re-considering iso-SPI transformer and Buck-Converter:

I added justifications for our choices to the [DR doc justifications](#) subtab

Next Steps: - Update the PCB layout with the changes and continue routing

CC: @Krish D

| Aarjav Jain (Feb 20)

| @Hemat Wander

Untitled

Author: Hemat Wander

Date: Feb 10

Implementing Auto-Connection Circuitry Schematic

From the [previous update](#), we decided to stick with using PFETs to enable the auto-connection circuitry. With that in mind, @Krish D raised the idea of trying to implement a way to have the balancing somehow before the auto-connection circuitry, that way we don't have to worry about the voltage drop over the PFETs during high current draw. However there are some complexities to this we dive into below. ** Option #1: **We have the balancing circuitry before **the auto-connection circuitry**

- As mentioned this would remove the high current draw through the auto-connection FETs

We need some way of pulling up the gate voltage of the PFET to stop balancing current from flowing.

- If we do something like this, and it doesn't work, it makes it harder to revert back to circuitry we are more confident will work. In other words, if we put the balancing circuitry FIRST, then we will **HAVE TO** pull up the balancing gates through some pull-up resistor. An example of such a pull-up is shown below.

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The issue with having such a pull-up is that it will passively charge up the entire ASIC through the balancing pin, meaning we have to rely on this ability to "charge up" to a high voltage through this high resistance (100k ohms) working. **We are concerned this might not work because we have never tested this before.**

Option #2: We make the balancing circuitry configurable to either connect BEFORE the autoconnection circuitry or to connect AFTER the auto-connection circuitry

- Same benefits as option #1

Allows for reconfigurability in case the pre-charging does not work, and we want to have the balancing circuitry strictly after the auto-connection circuitry

Will significantly increase routing complexity and take more space on board for components we use to choose which nets to short everything with.

Possible implementation of this: Using zero-ohm resistors or using 3-pos test pins which we short either way using jumper caps.

Option #3: Balancing circuitry is placed AFTER balancing circuitry. (this is the way it is placed on the schematic right now).

- Suffers from some voltage drop depending on the current from balancing. The specific amount could range from less than 100mV, to 200mV depending on the PFET's being used. - Don't have to deal with the above issues.

Conclusion: Considering the options above, I'm going to choose to go with Option #3 (aka keeping what we have). This is because: 1. It reduces complexity 2. It allows us to revert back to having other forms of auto-connection circuitry 3. If needed we can bypass the PFETs with hardware for improved accuracy if we find this is required from testing.

CC: @Krish D @Michael Lin

| **Aarjav Jain** (Feb 11)

| @Hemat Wander: Fully agree with continuing with Option 3.

Untitled

Author: Hemat Wander

Date: Feb 9

Coming to a decision on pre-charge circuitry: In the past month or so I have been re-evaluating the autoconnection circuitry a lot due to it being one of the main requirements we wanted to add for V4, and unfortunately finding that any

method I can find of doing so has somewhat major flaws.

Two sum it up, we essentially have two main options for satisfying this requirement. One is what I will call the PFET option, which effectively provides switches at each ADBMS1818 channel input, and the other is the pre-charge option, which involves charging up each ADBMS1818 channel input up to a similar voltage to the cell to protect against transient spikes when adding each connector.

**** Option #1: PFETs****

pros:

Have evidence that it works (from Formula E using effectively the same thing)

Would have a much simpler procedure for use. That being, we plug in the connectors in any order, and then just have **one** jumper cap that we need to plug into activate everything.

cons:

- Lots of space taken up on the board, expanding the board to 120mmx160mm, filled with components

tens to hundreds of milliohms of resistance almost guaranteed for any PFET chosen, making a 20mV-100mV level reading drop during balancing

Option #2: Pre-charge circuitry

pros: - Extremely simple and compact circuitry

- No resistance issues, meaning very little drop in voltage during balancing

cons:

- A more complicated connecting procedure. First we must connect the first and last module connectors and attach a jumper-cap for pre-charge. Once that is done, we can connect all other connectors in any order. Finally, we have to remember to remove the jumper-cap.

- Possibility it might not work (we have to test with an ADBMS1818 chip to confirm it works)

There is also a secret 3rd option, where we return to the same topology as V3 where we have jumper caps for each module. However, I think there would be a way to be able to at least connect the jumper caps in any order (by having a large resistance in parallel with the jumper caps to pre-charge).

So what should we choose?

Essentially, choosing between these two options comes down to prioritizing ease and simplicity of connecting everything, or effectiveness once everything is connected. Originally, I was leaning towards Option #2 as it would provide higher accuracy without the voltage drops across the source-drains of each PFET (especially during

balancing). However, after DR day and discussions with others, **I'm going to conclude that we should stick mostly with Option #1.** (I will get back to the "mostly" part)

My justification for this is that idiot-proofing the connection process is the most important thing, more so than having accurate readings during balancing. This is because, we want to create a procedure for working with the slave boards, which we can even follow when our brains are fried (after an all-nighter at competition). Furthermore, we shouldn't be balancing anyways during competition, so this lack of accuracy will only be an issue when using the battery outside of comp.

Note that this comes somewhat directly from the requirements in the DR0. I mean it was also a requirement to have accurate voltage readings, but I'm now choosing to prioritize this option.

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That being said...

I will also include the components on the board for option 2 (and option 3), which will only take up slightly more space on the board (8 more SMD components), but it will allow us to switch to an option without the balancing issue.

Some changes to the PFET circuitry

As mentioned in [previous updates](#), I realized that the gate capacitors would have rendered the circuitry completely useless by making the PFET conduct for a significant period of time when a connector is added. Today, I realized that the resistors in series with the diodes were effectively useless. I originally added them to have a more ordered connecting sequence for each module. However, that is not necessary, as we only need to ensure that the voltage at each input of the ADBMS1818 stays below 8V, and that there are no current spikes, both of which are true in this case.

One big thing is that I realized from looking at the PFET formula E is using, that there would be a voltage drop of up to 70V across the PFET when it is charged up. Thus, FE using this [PJA3471 R1 00001](#) PFET, which I can confirm after scouring digikey, is the best of its high voltage rating kind. The issue with it though, is that it has a much higher $R_{ds(on)}$ and much high minimum gate-source threshold voltage. Thus, for the goal of being able to stick with a smaller voltage rating PFET, I devised the following:

Use an extra 100k resistor in parallel with the PFET, to essentially charge up the drain of the PFET, such that the PFET doesn't have to handle taking a large ($>70V$) source-drain voltage. I did some spice simulations, and it seemed to work (results below). If it doesn't work, we should be able to substitute FE's PFET in-place of ours as they have relatively the same footprint (SOT-23-3 vs SOT-23), and we can de-populate the 100k resistors if needed. The resistor is 100k as to not blow up the ADBMS1818 during this pre-charge process.

Is this logic sound @Krish D ?

In the following simulations, we can see that the source-drain voltage of the PFET (this is for module 15) spikes up to ~15V for an extreme brief period of time before falling back down. Thus, the drain-source voltage stays within the limit and falls back down to ~0V extremely quickly. This is essentially a race for the drain to charge up sufficiently before the gate charges up, so as to prevent a large drain-source voltage buildup. There's a concern this might be a race condition, but if it doesn't work we can always revert to a more stable PFET.

[Missing image: ../images/image_2744685390.png]

Also, we are going to be switching to the [DMP2045U](#) for the PFET, as it is seemingly cheaper, has a lower Rds(on) and has ESD gate protection.

Next Steps: - Finalizing the changes to the auto-connection schematic with this in mind. - Changing the layout + routing to match these changes.

Aarjav Jain (Feb 11)

@Hemat Wander I like this plan!

Untitled

Author: Hemat Wander

Date: Feb 9

Pre-charge testing circuitry: The purpose of this update is to essentially experiment with one of our no longer in use slave boards and determine if I connected a pre-charge resistor ladder, what would happen to all of the voltages, or in other words, how much would the internal resistance of the ADBMS1818 affect the voltage ladder. ([Test plan here for reference](#)). As the test is pretty simple, the plan is pretty bare-bones.

Note although we are testing with the LTC6813, the application is for the ADBMS1818, which should be similar.

From a [previous update](#), I measured a resistance of 3.5 - 4 Megaohms from each ASIC input to GND (on one of the spare LTC6813s). Now, I'm going to try creating a test circuit to pre-charge the ADBMS1818 up to a reasonable voltage using high value resistors to prevent any current spikes.

I set up a chain of 100k resistors in series and then connected each one to a cell input, except for mod16, as the pre-charge relies on their being no direct connection to module 16. As I'm working with technically HV (60V) I taped all of the test points to the alligators and will try to only use one hand as best as possible. I also will set the current limit for the power supplies very low as we don't need much current. I also have the reverse voltage protection diodes for the PSUs connected as normal.

[Missing image: ../../images/image_2744413959.png]

I started up the PSU and started turning up the voltage of the 1st PSU while probing the second module. I got a voltage drop of 8V even when the pack was only at 22.5V. This shows clearly that there is some large imbalance internally with the resistances of the LTC6813. (At least with this possibly broken version.)

Actually, it was a hardware bug: I forgot to connect the GND of the slaveboard. Once I did that the voltage readings started being more normal (2nd module had 1V when the pack voltage was at 20V).

I increased the total series voltage up to 63V, and so now I'm ready to probe. Ideally, we would want the module voltages to be rising from 3.9375 V increments (however, the resistance of the ASIC might play a role + the ASIC might be damaged). Interestingly, we get voltage drops of ~0.5V across each module and like 56V across the top

resistor. (This is not something I expected, as we would expect the voltage across each resistor to be essentially the same). What this means is drawn out below.

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I then decided to probe the resistance measured across from each module to GND across the resistor ladder (while it was attached to the ADBMS1818) and I got essentially increasing resistances from 100k ohms to 1.6 Megaohms (as would be expected). Thus its extremely strange that we get such a voltage drop across the initial resistor.

My best guess is that it has to do with the LTC6813 on the slave board I'm testing with being broken. Either way, we can regard this setup as a failure, since it didn't work.

Conclusion: As this setup didn't work, we know that we would have to drive a resistor ladder with a lower resistance, meaning that instead of having the high resistance be on the ladder, the high resistance would be from the ladder to the ASIC (as in the drawing below). Note that the ladder is only connected to the most positive module and GND (this is different from the old setup).

[Missing image: ../../images/image_2744443639.png]

Untitled

Author: Hemat Wander

Date: Feb 6

Autoconnection Circuitry Testing #2: ([previous](#))

The purpose of this update is to test the two methods for auto-connection testing and see some of the behavior I missed last time. Previous update for context. Specifically I want to:

- Repeating Tests with the fix to the breadboard circuit **with** the PFET gate capacitors. (I had messed up some resistor values previously)

- Try tests with now removing the gate capacitors (as we are planning to in the main design).

- Probe balancing voltage drop when drawing current across a higher module (say module 5)

Assembling Test Components: The first thing we need, as in the previous update is diodes to protect the power supplies when connecting them in series.

From the inventory I used the following diodes:

- 2 slave board diodes that were in the proto-crate

- Also used 1 through hole diode that was left over from the testing I was doing

side note: I realized that soldering on top of old stencils makes it a lot easier (so we don't damage the desk)

side note #2: I probed continuity across the power supplies when they were off (

This gave me the six diodes I needed for all of the power supplies.

For the test plans below, I [referred to this doc](#).

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Test Setup #1: Completing the original test from before

This test entails actually connecting the autoconnection switch and probing with an oscilloscope to see what happens. Thus, we first connect all of the power supplies in series, turn them to 4V and connect them to the breadboard as before. We can then probe the voltages across the 5th and 6th module to start. Below are two photos taken from the modules being connected, we can see that their voltage seems to spike up to ~8V at the beginning. (A zoomed out and zoomed in photo are shown). **The red line shows the difference in voltage between input 6 and 5 (which is effectively the voltage of mod 6). **

Module 5 and Module 6:

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This is unfortunate, because this is the opposite of what I wanted to see. I was expecting to see the voltage curves be much smoother and more in order. It seems that changing the resistance values from 100ks to something else as I simulated should make it better, actually made it worse in real life. (I re-tested this a few times and got similar results). Its possible that something with the power supplies transients is what's causing this, but I'm honestly not sure.

Below are some of the other modules, they all seem to have better voltage curves (none of them spike to 8V), but nothing is charging up in order like I was expecting. I have no idea what's causing this.

Module 4 and Module 5:

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Module 3 and Module 4:

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Module 1 and Module 2:

[Missing image: ../images/image_2740344645.png]

****Module 2 and Module 3: ****

[Missing image: ../images/image_2740345109.png]

Interestingly, module 1 and 2 and then module 2 and 3 seem to display the exact behavior I was hoping for where their voltage curves rise up one after another, where they charge up one after the other. It's very strange that none of the other ones are working other than the first two.

Conclusion: Either this is an issue with my breadboard (which I could not debug), or something is stopping the circuit from auto connecting correctly. Either way, this this output not working is fine due to the outcome of the test without a capacitor (see below). (from later results I am guessing this is due to a hardware bug?)

Test Setup #2: Connecting Modules out of order (WITH PFET GATE CAPACITOR)

The purpose of this task is to simulate plugging in the connectors on the slave board out of order before we get to the pseudo-steady state of all the connectors being plugged in.

Before beginning the test proper, I tried testing what I would see if I connected say module 5's jumper wire without connecting anything else (except GND and module 1). If I probe the other side with the orange multimeter I see that the voltage spikes up quickly before going back down.

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Now probing with an oscilloscope.

****Connecting Module 1 + GND and then module 5: ****

If we plug in this module, with mod 1 and GND connected, we see that the voltage spikes up and then falls to around 4V where it seems to oscillate around for a few seconds before falling back down to 0V. This oscillation behavior is a bit strange, but the sitting at 4V before falling down is expected.

[Missing image: ../../images/image_2740348170.png]

[Missing image: ../../images/image_2740348793.png]

[Missing image: ../../images/image_2740349491.png]

(All of the above photos are of the same thing just zoomed differently).

****Connecting module 5 and then module 1 + GND: ****

We seem to get the same behaviour as above.

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Connecting Module 1 + GND and then module 6:

The difference between this test and the last two is that module 6 is connected to the beginning of the gate PFET pull-up chain. So this simulates connecting the first and last module. We seem to just get an even bigger spike in voltage.

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I don't think there is a point testing the other modules with capacitors. Now, we can test everything else without capacitors, as we decided based on simulations that we would be removing the capacitors. (The preliminary tests enforce that).

Removing Gate Capacitors:

As can be seen from the schematic below, the capacitors I am removing are the 100nF capacitors on the gate source of the PFETs. These are currently preventing the gates of the PFETs from being pulled up quickly when a "connector" is attached.

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Test Setup #3: Connecting Modules out of order (WITHOUT PFET GATE CAPACITORS)

Same as before, but we now have capacitors removed. We get a similar spike at the input, but this time the spike is for much less time than it was for before + we no longer see the oscillatory behavior at 4V. (For mod 3, we see some oscillation but its at a much smaller voltage).

Module 6:

[Missing image: ../../images/image_2740355197.png]

****Module 5: ****

[Missing image: ../../images/image_2740360332.png]

Module 4:

[Missing image: ../../images/image_2740360606.png]

Module 3:

[Missing image: ../../images/image_2740360923.png]

Conclusion: Even though a voltage spike is still present, it fades to 0 much more quickly, and is much less pronounced. This voltage spike is caused by the time it takes to fill up the PFET gate capacitance before it can stop the flow of current. If we need to, post-bringup we can reduce this resistance value (along with the others to compensate at the cost of higher quiescent current). Furthermore, I'm also thinking of using PFET diodes with builtin TVS ESD protection, that will hopefully allow some current to flow from source to gate during transients to help flatten this curve.

Test Setup #4: Same as test Setup #1 but with capacitors removed

When using 50ms time intervals, module 5 and 6 now seem like they are being connected at the same time. However, if we zoom in to 250us, then we see much clearer behavior. The modules seem to be connecting one after the other!! (this is even better than with a capacitor). **Reminder:** Red is the second module (through math), and yellow is the first module.

Module 5 and 6:

[Missing image: ../../images/image_2740361626.png]

[Missing image: ../../images/image_2740361889.png]

Module 4 and 5:

[Missing image: ../../images/image_2740362230.png]

Module 3 and 4:

[Missing image: ../../images/image_2740362620.png]

Module 2 and 3:

[Missing image: ../../images/image_2740363116.png]

Module 1 and 2:

[Missing image: ../../images/image_2740363530.png]

Conclusion: The autoconnection circuitry works better without capacitors. However, the fact that module 3 didn't seem to work, possible points to the fact that it was indeed a hardware bug that stopped the testing **with** capacitors from working. Regardless, this is good for without capacitors.

Test Setup #5: Testing balancing voltage drop with capacitors removed

The purpose of this test is to see the drain-source voltage drop across the PFETs when we are drawing more current due to balancing. I specifically realized I wanted to test this at higher modules, mod 5 and 6. I also ensured I was not hitting the current limit of the PSUs for the following tests.

To give some context, the gate source voltage and drain source voltage locations for the PFETs are shown below. If we were to be balancing module two, then the current "I" would be flowing as shown down through PFET 2 and up through PFET 1.

[Missing image: ../../images/image_2740367170.png]

Mod 6 balancing

We see a voltage drop of 160mV across the PFET of the 6th input and 60mV across the PFET of the 5th input. This creates a total drop from 4V to 3.5V across the balancing resistor, however this is likely due to additional resistances in the breadboard. the voltage drop across the PFET is the main store, and this level of voltage drop is unacceptable.

[Missing image: ../../images/image_2740364266.png]

For reference, the gate-source voltage of the 6th PFET is -3.3V and the gate-source voltage of the 5th PFET is -3.4V. These should be plenty for balancing, so it **extremely strange** that we are seeing this level of voltage drop. It could be due to the power supply somehow?

[Missing image: ../../images/image_2740365013.png]

Mod 3 balancing I tried to test mod 3 to see if it was an issue with the power supply, but I got that the voltage drop across PFET 3 was still 50mV. For reference, the gate source voltage of the 3rd PFET was around -3.4V. The voltage drop across the PFET of mod 2 was in the 30mV range (as last time). This is with a -3.57V gate-source voltage drop. It just doesn't make sense that we would see such a large change from 30mV to 50mV+ just from the small change in gate-source voltage.

If we do the same as above, but drop all of the PSU voltages to 3.0V, we get a gate-source voltage across the 3rd PFET of 2.5V, and then the voltage drop across the drain-source of the PFET becomes 22mV. This demonstrates to me that the voltage drop across the PFETs is not an issue of gate-source voltage not being sufficiently high.

If we try the **Mod 6** balancing test, but with 3.0V PSUs instead of 4.0V PSUs, we get that the drain source voltage drop across the 6th PFET is 45mV with a gate-source voltage of -2.5V. I guess this discrepancy is just caused by the PFETs having different gate source voltages, however I was not expecting them to vary **this much**.

****Switching the PFETs and then re-testing. ****

The purpose of this test is to see if the difference in voltage drops is due to the variation of the PFETs. After switching the 2nd and 6th PFET I decided to retry the balancing on module 6 (drain-source voltage drop across the 6th PFET when the PSUs were at 3.0V). I got a drain source voltage of 35mV compared to 45mV as before.

If I set the PSUs to 4.0V each, then I get a drain-source voltage drop across the 6th PFET of 39mV compared to the 160mV from before (big difference).

If I do balancing across the 2nd module and probe the 2nd PFET I still get a drain-source voltage drop of ~28mV still. So it seems to be a combination of the module # used and of the PFET specifics. That, or something is going wrong with my breadboard circuit at these highish currents (~0.3A).

****Conclusion: ****

The voltage drop across the PFETs seems to vary a lot, and so if we value retaining voltage information while balancing, it doesn't make sense to me to use PFETs. If we are fine with only reading voltage information while balancing, then this is fine. Otherwise, we will likely need to use the pre-charge circuitry instead (if it works that is).

Overall Thoughts:

Based on the above results, it seems that although we were able to get very nice behavior with the PFETs switching on in order (without gate capacitors), there unfortunately is the issue of the major voltage drop during balancing. Based on how much it seems to vary, we would likely have to find some alternative solution to read voltage during balancing (i.e. PWM balancing). What are your thoughts on this @Michael Lin @Krish D ?

One thing is that we are using 10Mega ohm pullups for our PFET gates, while formula E is using 100k pull-ups (100x difference). This means our spike might be much longer than theirs would depending on the PFETs they used. On their schematic I don't think it says the MPN of the PFETs they used, can we ask @Krish D ?

Another final note is that the voltage drop probably only varies for forward current through the PFETs. Backwards current through the PFETs would always have a similar voltage drop from the body diode.

Next Steps:

- Test the pre-charge circuitry using the spare slave board and measure the voltages across each cell input with a 100k resistor divider ladder for pre-charging. - From the above, decide what autoconnection method we want to use.

Christopher Kalitin (Feb 6)

@Hemat Wander

If "PWM Balancing" how much time do you think would be required after stopping balancing to taking the voltage measurement?

I imagine this becomes an issue at very high voltage measuring rate. Eg. If you need 1 ms delay to get a good reading, but take readings at 500 Hz then your effective balancing duty cycle is 50% (each period is 2 ms, half of which has the PFET off for you to take a reading).

Also you should ask the people who designed Formula E's boards directly, the equivalent of FE asking Hanlon questions. Or call Hanlon or other previous solar Goats!

Untitled

Author: Hemat Wander

Date: Feb 5

The purpose of this update is to document a discussion on slack that happened a while ago.

Slave board Positioning and Accessibility:

The following drawings show what the wire accessibility for the scrutineering and isoSPI connectors would be from the pack. As a note, the scrutineering connector would be tied down somehow using some kind of removable holder (maybe velcro?).

[Missing image: ../../images/image_2738774472.png]

Aarjav Jain (Feb 6)

@Hemat Wander @Deev Shah @Krish D : Hey Velcro is not a bad idea for this application actually.

Deev Shah (Feb 6)

I agree, velcro is a good idea for a quick and easy fix

Untitled

Author: Hemat Wander

Date: Feb 2

Auto Connection Circuitry Rehaul #2: Today I probed the resistance of one of the spare LTC6813s that we had in the BMS crate, because I wanted to see how much the ASIC was going to load the circuit, and therefore how low we need to make the resistance of the resistor divider. Note: I used the big digital multimeter, as it seems to be much better with larger resistance values.

Probing Results:

When I probed between adjacent pins to try and find the internal resistance of the ASIC between them (Ex. C18 and C17), I found that the resistance was in the range of 3.5 - 4 Megaohms, although the value seemed to shift around.

- When I probed between pins spaced out by 1 (Ex. C18 and C16) I got an internal resistance of 50+ Megaohms.
- When I probed between each of the pins (C1, C2 ... C18) and V- (GND), I got an internal resistance of only 3.5-4 Megaohms.

Note these might be dependent on the specifics of the voltage/current output of the multimeter, and so we will have to test with a real circuit to determine them.

[Missing image: ../images/image_2725121522.png]

Simulating in LTspice As from above, it seemed that the 4 Megaohm isolation to GND for every pin would be the limiting factor in this case of loading the resistor divider ladder I was making. (loading meaning it would draw current away from the ladder, making it not have a perfect series of voltages). Thus, I tried simulating in LTspice with a 4 Megaohm resistance to GND.

[Missing image: ../images/image_2725120508.png]

Essentially, we first charge up all of the modules using a resistor ladder to GND. However, since the ASIC itself has a non-negligible resistance, it pulls every point to GND across this large 3.5Megaohm - 4Megaohm resistance. See the below simplified diagram for reference. We first connect the top module to the resistor divider, and then connect all of the modules once the ASIC has been "charged up". This prevents any current spike internally in the ADBMS1818.

[Missing image: ../images/image_2729234344.png]

The "issue" is that the internal 4Megaohm insulation to GND at every point, makes the resistor divider, no longer a perfect resistor divider. Thus, every voltage (C1, C2... C17) gets charged up to some voltage slightly below its actual cell voltage. For example, module 17 gets charged up to ~52V while its actual voltage is ~60V when plugged in (if the cells were at 4.0V).

For these simulations for example, I switched the C17 switch first AFTER already recharging with the resistor ladder. The following are the current spikes we see internally in the zener diodes of the ADBMS1818. As a reminder, the max current that can go into the ADBMS1818 pins is 10mA.

[Missing image: ../images/image_2729219292.png]

However, these are still smaller than the current spikes I got from FE's circuitry when connected module connectors out of order (which we know should work in theory). This leads me to believe it will be fine.

The other point that

made was discussing the quiescent current draw of this resistor ladder. It will essentially always be connected, so we can do a worst case calculation. From simulation, I got a 100 uA - 200 uA draw from each module. If we assume 3.5Ah per cell, then we get:

$(3.5 * 13) / (200 \text{ uA}) = 227500$ hours to drain the battery.

Alternatively, we can view it as the power loss during the race (let's say 3 days non-stop).

$72 \text{ hours} * 3600 \text{ seconds} * 200 \text{ uA} * 4.2\text{V} = 218 \text{ Joules}$.

I couldn't find from the strategy wiki page what number of laps this would correspond to.

If you can find it.

Note this quiescent current draw is similar to the 100k resistor divider for the PFET autoconnection circuitry as well.

- I can try playing around with the resistor ladder values, to try and calculate what resistances will get us closer to the voltage of each module (so that way it is not that far below).

We should try and find the extra slave board to see if we can connect a pre-charge ladder to it, and see what voltage the modules will rise to. If we find it, we can conduct tests in the next week.

I can try completing the IRL autoconnection circuitry tests with the breadboard (with and without gate capacitors), which I originally delayed because I thought the testing wasn't going to be necessary anymore.

Based on the results of the above, I will either add both the PFET and charge up circuitry, or only one of the above.

Untitled

Author: Hemat Wander

Date: Jan 30

Rethinking Auto Connection Circuitry: When I was thinking in the shower a few days ago, I realized a problem with my circuit which I had never considered to actually test in simulation before. Essentially, all of my simulations (and real life tests so far) were focused on how the auto connection circuitry would behave when all the connectors have already been plugged in and it was in a pseudo steady-state. Thus all my simulations were concerned with if the PFETs would properly begin conducting in order. However, I never properly considered what would happen when the module connectors are actually going to be plugged in to get to that steady state.

I have attached all relevant spice files I used for simulation.

The Problem: Essentially, the PFET will begin conducting for a short period of time if we just plug one of the modules in. This means that if we connect the module board connectors for modules 8 and 9, the PFET will conduct for a short while **because** the gate voltage has to charge up to the drain voltage. I originally thought the 10M resistor would pull up the gate immediately, but the large resistance in combination with the gate capacitance + the other capacitance makes it charge up more slowly. Reminder: We have these large resistances to reduce the quiescent current draw.

[Missing image: ../../images/image_2720596420.png]

We can simulate what happens when the autoconnection switch is open, and we plug in a random connector, say Module 15's connector. Reminder that the purpose of this circuitry was to be able to connect the minifit connectors in any order.

[Missing image: ../../images/image_2724372447.png]

When we simulate this, we find that there is a 50mA spike across the internal Zener diodes of the ASIC when we connect the circuitry, which could cause the ASIC to break (the limit given in the datasheet is 10mA). This spike is caused by the PFET being in conducting mode for a long time (in the seconds) before the gate gets pulled up, when the PFETs should really be blocking them from the ASIC.

[Missing image: ../../images/image_2724374091.png]

[Missing image: ../../images/image_2724407676.png]

What about Formula E's Circuit? My first thought after seeing this was wondering if Formula E's circuitry also had this problem. If we simulate, we see that the voltage across the filtering capacitors spikes quickly before returning back to zero. Formula E's circuit is able to do this because the lower resistance gate pull up resistors are able to more quickly turn off the PFETs. We also still see the current spike across the internal Zener's go up to 20mA as well.

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[Missing image: ../../images/image_2724394037.png]

So you might be thinking, we can just use Formula E's circuitry, and we probably could since we know they have tested it and it works. However I'm weary to do so for a few reasons. - As they are using PFETs to connect their voltage taps any time we draw high current (balancing) there will be some voltage drop in the magnitude of ~50 mV depending on the PFETs used. (I know this is also true for our, but I address this later down). - The spikes I see in simulation for the voltage and current make me cautious about the chance it has to break the ASIC, especially in high stress conditions. - The quiescent current draw of their circuit in the mA range, means that if the circuit was left plugged in for about a year, it would discharge 20%. This is perhaps a lesser reason. $(24 \text{ hr} * 365 \text{ days} * \sim 1 \text{ mA}) / (3500 \text{ mAh} * 13 \text{ cells}) = 19.3\%$

NOTE: These are pretty small reasons, so we can come back to their circuit if need be. Also, our original circuit might still work, depending on how wrong the simulations are. We could test to see if the IRL autoconnection circuit also faces these problems. * Side Note: [*Previously](#), I simulated formula E's circuitry, however I know see that I simulated it wrong because I had included capacitors when there were any. If I simulate it now, I get the following results with the connection order, where everything connects in around 0.1 ms. (However, the voltage curves still seem a bit messy, and according to simulation there is a 20 mA spike in the internal Zener diodes which should break the chip). Formula

E's circuit probably works however since the the current spike happens for such a brief duration.

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Attempts at solving the problem: When I found that this circuitry might not work, it kind of worried me, but in the spirit of re-thinking BMS from the bottom up, I decided to do some more brainstorming to try and come up with different solutions than Formula E. Below are some of the attempts I made.

TWO POSSIBLE SOLUTIONS: After about a day and a morning of brainstorming (and some LLM help), I came up with the following two circuits we can use for pre-charge.

1. The first idea is to use high-value resistances to pre-charge the modules before connecting the actual module connections. Essentially, the requirement is that we connect the first and last modules connector first (meaning we would have to have to ensure we at least plug in the last connector first before connecting anything else). Then, the highest module voltage forms a voltage divider down to the ground of the circuit through the use of 10Mega ohm resistors. This effectively pre-charges all of the ASIC inputs up to the correct voltage. After pre-charge, we can connect the jumper cap and then connect the rest of the connectors in any order we would want.

possible issues:

2. IFF the above circuit doesn't work, we can try a similar method of pre-charging, but in this case using PFETs to pre-charge instead of connecting the pre-charge all of the time. With this method, we could likely solve most of the issues we **could** have with the above circuit depending on how testing goes. I'm not exactly sure about the details of how this circuit would look, but I will look into that if only necessary.

Conclusion:

By using a pre-charge approach as opposed to a 1 by 1 connection approach, we have a better chance of having a functioning circuit and no longer require PFETs to be constantly connected (meaning we [don't have issues of voltage drops during balancing anymore](#)). AGAIN, I have no idea of if this will work, other than it seems to fix the issues of current and voltage being seen by the ASIC (we will need to test to determine this). The main cost is that we now NEED TO ENSURE that the first and last modules are plugged in first, however I think that cost is justified.

The plan is to use the 1st solution, and only to consider the 2nd solution if we find from testing that the first solution doesn't work. If the first solution works, we might be able to remove a lot of the auto connection components we have on the board.

What do you guys think @Krish D @Michael Lin (this will probably extend slave board making by a bit, but I think it is a better solution, and has a better chance at working).

Contingencies:

If none of the above options seem good (or we don't have time for them in our timelines), we can try using the same circuitry we are now, except with no capacitor. The voltage spikes and current spikes look about the same as FE's circuit, so it should in theory work.

Next Steps:

The first thing I want to do (if I can) is to take the slave boards out of the pack, and then try pre-charging one using a chain of resistors. This will consist of using two power supplies in series (60V total), and then connecting that across a ladder of resistors (we can decide the value, perhaps 100k ohms - 10 Mega ohms). Then, we can probe the voltages across the ASIC to see if it is working.

- Another thing I could do if the first idea isn't an option is to probe isolation between the LTC6813 pins and GND for some of the spare LTC6813's. (The LTC6813 will behave similarly to the ADBMS1818).

Reflection: Before I begin this reflection, I want to be clear that we have not tested these pre-charge methods, and so we won't know if they work. That being said even in terms of just consider this option: This is quite a big overhaul to the existing work I was doing on the slave board circuitry and this option probably could have been evaluated earlier in the process. Why wasn't it? In my mind, I was strictly thinking of the requirement for the slave board being that we need some method for connecting each of the ASIC inputs in order. However, that's not what the requirement was listed as in the DR. (See image).

[Missing image: ../images/image_2724544116.png]

Even in the beginning of the process, I was already thinking about ways of switching on each connector in order. Instead, I should have been thinking of how to connect the ASIC without breaking it. Breaking it in this case means trying to avoid frying the chip via high current spikes and breaking the voltage thresholds. I could have avoided this simply by thinking bigger earlier on, and not getting stuck in the details of trying to solve a problem that wasn't the actual problem.

The thing is, even before the slave board DR0 happened, even before competition in 2025, I had already subconsciously made the requirement in my head that there needs to be some way of connecting the inputs in order. Then when I saw Formula E's schematic doing that, I got further locked into this one solution space.

To not have this problem in the future, I essentially just need to reconsider if what I'm doing is actually tackling the core of the requirement, or if I made some assumption about how the requirement needs to be solved. This can be done by questioning yourself IN THE EARLY STAGES, for if the way your considering to solve a problem, is the only way that problem can be solved.

Extras:

In this process of brainstorming, chatGPT told me this... so I probably should have asked chatGPT this question earlier on.

[Missing image: ../images/image_2721528180.png]

Also chat gave me some useful links that could be helpful later on, for some forums and a datasheet for an application of the LTC6813:

[Where to attach shield of isoSPI ADBMS1818 vs LTC6813](#)

[Mixing ADBMS1818 and LTC6813 in one daisy chain](#)[List of possibly useful threads](#)[Other ASIC thing that we might want inspiration from](#)

LTspice Files:

[Auto Connect No Capacitor.asc](#)[Auto Connect No Capacitor CONNECTING.asc](#)[Auto Connect Input Capacitor.asc](#)[Auto Connect Input Capacitor CONNECTING.asc](#)

Krish D (Feb 1)

@Hemat Wander Amazing work analyzing your prospective outcomes and questioning the exact nature of your solution against the requirements. This the **exact** mindset you should have during the design stage of a PCB project.

Regarding your technical points, I have some notes that led to some questions, which I'll present below:

1. The 100k resistor divider ladder:

Balancing issues: As mentioned from your points about quiescent current draw, this solution is that you will always be balancing your modules. To determine the significance of the issue, you should find the absolute highest power loss as a result of continuous voltage difference between modules, and compare that against the

Relying on procedure: Original problem of V3 was that it wasn't obvious that too not disconnect them in any order, so your previous solutions all relied on **a single jumper** to do all the work. This solution somewhat fails to address this since it relies on having to plug-in and unplug connectors in a defined order. However, this might be a reasonable tradeoff since plugging only 2 connectors in/out in a particular order is much easier than 16.

This solution is less bulkier than the existing autoconnection circuitry and can be tested with our existing slaveboard, however how feasible is it from our timelines perspective to test this and not create an issues with existing slaveboard that we want to use for driving day? The largest concern with testing this is that we could break the LTC again, and with our current limited supply, this could prevent us from having the pack in drivable condition, and could also prove to be a major debug time sink.

You mention that having the PFETs is a cause of concern due to voltage drop across the R_{ds} on of the FET during balancing, however this is only significant if we are to actually balancing during comp.

2. Removing the capacitor before the PFETs:

- There is no way to test the feasibility of this solution unless you bring-up the new slaveboards, as it would be quite difficult to adapt your breadboard circuitry to see if this will damage the ADBMS1818 or not.

Conclusion: What are your thoughts on the points above?

The priority should be to determine what solution matches all of our existing requirements, and to be able to test them for feasibility. Below is the suggested course of actions to ensure we can fail fast, and achieve an MVP for the slave boards ASAP, without risking the integrity of the slave boards currently in the pack.

Increase the size of the V4 Slaveboards as needed to add space for the 100k resistor ladder, HOWEVER, leave these as DNP for now.

Assuming schematic is ready otherwise, order the Slaveboards.

While waiting for order to arrive, look for our **extra** slaveboard, and use a 100k resistor ladder divider on a breadboard to test if it is feasible to connect the existing slaveboard in out of order in the method your originally proposed.

Note: For step 3, consider that we currently need more LTC or ADBMS1818 units to use in the event that one of them breaks during testing.

4. When the slaveboard arrives, if the previous solution hasn't been tested on the extra slaveboard, than firstly, test the 100k resistor ladder divider solution. <- (Consider what exactly needs to be measured and how this going to be done).

5. Next, remove the 100k resistor divider and test the auto-connection circuitry as is/was. This includes removing the capacitor at the gate of the PFET to see if this is still feasible (the definition of feasible here means the modules can be plugged in any order, the autoconnection circuitry works as is.

From here, feasibility of either of the solutions can be assessed all at once. Also consider than you can test the different solutions in parallel by bringing-up a second slaveboard in parallel.

Please let me know your thoughts @Hemat Wander

CC: @Aarjav Jain

Untitled

Author: Hemat Wander

Date: Jan 26

Testing Auto Connection Circuitry Breadboard Circuit:

First, I soldered a few diodes to protect the PWS2323 power supplies, these diodes were connected in parallel with the supplies, with the anode facing the negative end. There weren't enough diodes to protect the other supplies, and I don't think right now it is worth it to look for some diodes ass we are only working with 4V per supply.

I wired up the supplies as follows, specifically with them being wires in series, and each positive end for each PSU being connected to a PFET source. Finally I connected the GND end. All of the exposed connections were taped up. I did a final check to see that no modules were continuous to each other to prevent shorts.

[Missing image: ../images/image_2708898347.png]

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I then also checked to confirm nothing on the breadboard was touching each other that it shouldn't (moved things apart, check by inspection).

Now I set all the voltages to 4.0V for each of the supplies with the auto connect switch disconnected. Probing across the first capacitor, we get 0V for the voltage. (this is what we expect).

Now we can connect the auto-connection switch using a yellow jumper wire. And we see that the output capacitor now reads 3.99 volts (this is good!) Similarly, we can probe the rest of the output capacitors when the switch is connected

and disconnected. We see that when connected, the output voltage is 3.99V - 4.03V while it becomes 0V when the switch is disconnected.

Something strange is for the last 3 voltage sources, I found that the voltages were a little more unstable (shifting around a lot (by $\sim 0.01V$), although I don't know if this is because of the circuit, or because the last 3 PSUs were from the XT 30-2 power supplies, which may just be more unstable).

Since I don't have much time, I will just be testing a few modules with the oscilloscope.

When I started trying to probe, I found some weird behavior with the 100 ohm resistors for the RC filters having a 2V drop across them, I'm extremely unsure of what can be causing this.

Ok, I found the issue, somehow the oscilloscope probes were acting as a short between the capacitor outputs, so I'm unsure on how to fix that. Currently, it seems like the GNDs of the oscilloscope channels are connected so I cannot use two channels at the same time. For now, we can just make the GNDs go to the same place, so the final voltages will rise at the same time.

OKOK after playing around with the oscilloscope for a bit, I finally got it to acquire a single shot waveform, by pressing the single button. (That shouldn't have taken that long.) We got the following waveform.

[Missing image: ../images/image_2708899783.png]

Side Note: When we disconnect the auto-connection circuitry, it takes about 5 seconds for the capacitors to become discharged. This is about what we expected, but is interesting to confirm.

Now we can compare the voltage charge time of the second and third module. The captured waveform is shown below. Blue represents the total voltage of the third module compared to the first module (C3-C1), while yellow represents the total voltage of the second module (C2-C1). The red line represents their difference (C3-C2) and therefore the voltage across module 3. We would expect this to charge up after the second module (C2-C1), but it charges up at the same time. It seems that the rise times of the capacitors seem to be exactly the same somehow. This is quite surprising to me and unfortunately means the capacitors are not rising one after the other (I was expecting at least a 1ms delay between charging). However, them rising at the same time will probably also still work for the connection to the ADBMS1818, so this is not an issue really. Also it is nice to see that the voltage charge up is so stable.

[Missing image: ../images/image_2708900141.png]

If we compare the first and third module, we see that they also charge up at the same time, similar to the second and third modules. It seems that the charging will occur somewhat simultaneously no matter what.

[Missing image: ../images/image_2708900596.png]

Interestingly, on this third attempt, I tried to probe 3 channels, at module #2,

3 and #4. I found that if we probe the module voltage curves, we

get this interesting double bump for the voltages charging up. I'm not entirely sure what's causing this, as we don't see this in simulation.

[Missing image: ../../images/image_2708900958.png]

I also tried recording the falling voltages during module disconnection, and I found that they again seem to be falling at the same time rather than one after another. See below.

[Missing image: ../../images/image_2708901431.png]

Lastly, I will try to probe the gate source voltages for neighboring FETs. Now this is very strange, we see that the first module's PFET gate voltage is going down while the rest of the module PFET gates seem to be remaining at the same value. I actually am not sure why this is, and it is probably why we are seeing the output voltages behaving so strangely.

[Missing image: ../../images/image_2708902551.png]

To debug this, I probed voltage across the first diode and found that there was a positive voltage drop across it. OH WAIT, this is because the oscilloscope is essentially by passing the 10M resistor as it only has 1M of isolation. So unfortunately, we cannot really probe the gates as it stands. Thus this diagram is essentially meaningless.

Final test: current draw. The last test I wanted to do does not require the oscilloscope, only a multimeter to read voltage across the 10 ohm resistor. Across the second module. Before this "balancing resistor was added, the voltage drop across the capacitor was 4.04V. After connecting the resistor, the voltage we are reading becomes ~3.7V however, the voltage drop (source-drain) across the PFET only goes up from 0.04mV to 26mV as expected.

I tried decreasing all of the PSU voltages and repeated this test I got the following PFET drain source voltage drops.

$V(\text{PSU}) = 3.5\text{V}$, $V(\text{PFET}) = 26\text{ mV}$.

$V(\text{PSU}) = 3.0\text{V}$, $V(\text{PFET}) = 27\text{ mV}$.

$V(\text{PSU}) = 2.8\text{V}$, $V(\text{PFET}) = 25\text{ mV}$.

$V(\text{PSU}) = 2.3\text{V}$, $V(\text{PFET}) = 23\text{ mV}$.

$V(\text{PSU}) = 1.5\text{V}$, $V(\text{PFET}) = 36\text{ mV}$.

These results are much better than I expected, and essentially show that the voltage drop across the PFET even at low module voltages will be negligible for balancing.

[Missing image: ../../images/image_2708903676.png]

Conclusion:

Although the transient (when we connect and disconnect) curves did not behave as we expected in terms of one voltage input turning on after another (on the scale of milliseconds), we did see that all of the modules do seem to turn on at essentially the same time, meaning that we will never see any voltage across the ASIC inputs that would cause it to break. Thus, this circuitry would work if we use it.

- The voltage drop across the PFET is even less of a concern than I thought it was going to be, so it should be completely fine.

Aarjav Jain (Jan 26)

Excellent work @Hemat Wander. Your dedication especially these last weeks have been amazing! Writing this update during and right after your testing to inform us what is going on is extremely helpful. Some notes regarding the update:

In general one improvement to this update would be clarity. Whether you are writing this at 4am dead tired or fully energized, your update quality cannot decrease. This is because an update is only as good as people understanding it! For example you said "To debug this, I probed voltage across the first diode and found that there was a positive voltage drop across it". But it was not clear why you decided to probe this exactly. Perhaps a diagram in the update showing exactly what is where will help (or using Designators from Altium).

Hemat Wander (Jan 26)

I will respond to your points later, for now I wanted to note that I realized a mistake with the circuit setup that might resolve some of the problems I faced.

In the circuit, I forgot to place 220k resistor and 470k resistors respectively in series with the last 2 module's diodes. See [schematic here](#). This is likely what is causing all of the modules to charge up at the same time rather than one after another. Thus, immediate next step is to change these resistance values and re-test.

I also realized I need to test balancing voltage drops on the later diodes, as I was only testing with the second module. I.e. I only connected a 10 ohm resistor between module 2 and module 1, rather than testing between module 6 and module 5 for example.

POSSIBLE ISSUE: Something I just realized is that the auto connection PFETs will have a voltage drop of ~30mV across them (from source to drain) when balancing is occurring. Originally, I thought this value was fine, as I was mostly concerned with the voltage drop across the PFET changing the amount of balancing current (by adding the PFETs series resistance in series with the 10 ohm resistor).

However, I now realize that a 30mV drop across both PFETs will likely represent a 60mV drop in the voltage reading when we are balancing. AKA, a module reading 4.03V might become 3.97V when balancing. This could be problematic because balancing occurs for differences of 50mV. This may or may not be acceptable depending on a few points.

- First of all, formula E must be experiencing a similar thing, although I'm unsure what specific PFET they are using, as it doesn't say in their schematic.
- We might be able to find a smaller voltage drop PFET, but I doubt many PFETs will go below a ~10 mV drop.
- We could have a firmware solution for this (@Michael Lin) by compensating for the voltage drop when balancing (based on the expected actual module voltage). Although this seems like a band-aid solution, I feel like it might be our best bet, as we will be seeing a voltage drop across the PFETs no matter what.
- This depends on the characteristics of the cells, but another firmware solution we could try something like PWM discharging, where we balance for some 90% of the time, and then the other 10% of the time we stop balancing to again read the module voltage.

Aarjav Jain (Jan 26)

@Hemat Wander:

Untitled

Author: Aarjav Jain - Electrical Director

Date: Jan 25

Summary Current Design

Goal: Confirm that the current status of the auto-connection circuitry is feasible to manufacture and have confidence the current design will be able to meet its requirements.

TL:DR: We are confident. We have many justifications all coming from distinct sources which support our confidence.

****Requirements: ****

Confirmation: Below I list **design justifications** for the auto-connection circuitry. These justifications give us confidence that the circuitry **will** meet the above requirements:

CC: @Krish D

Untitled

Author: Hemat Wander

Date: Jan 23

Assembling Auto Connection Circuitry Breadboard: The plan for today is going to be assembling the slave board's auto-connection circuitry [outlined here](#).

Other than the materials outlined in the doc, I also need:

Components used: - 43 header pins

6 SOT-23 breakout boards

6 DMP2110U-7 FETs

[Missing image: ../../images/image_2706888001.png]

Some notes from assembly:

The method I used for soldering was quite inefficient and produced solder blobs on the PFETs. Next time, we can improve the process by using more flux and using a heat gun to melt all the solder under the component to flow the component into place.

- I tried measuring the resistance of the 10M resistors using the orange multimeter, but when I measured it, it says it was ~4.7M ohms. Now why can this be? @Gurman Khella However, when I measured it using the digital multimeters we had, it says it was 9.7M ohms, which seems closer to what we expect. If the multimeter is around 100M ohms, we can assume the resistors are basically 10M.

Also side note I measure the resistance of the breadboard and across 12V and GND on the new DRD and I got >>100M ohms on the breadboard, and 2Mega ohms across the DRD 12V and GND.

The final result:

[Missing image: ../../images/image_2706887634.png]

Untitled

Author: Hemat Wander

Date: Jan 18

Auto Connection Update #7? It's been a while since the last update. I thought I was done looking at this autoconnection circuitry stuff, but there are two last things I wanted to do before [testing the circuitry](#). One is that I realized a concern with the continuous gate-source voltage value of the PFETs, and the other is that I wanted to simulate formula E's circuitry to see what it behaves like (I never did this although I should have).

Concern for continuous-on PFET resistance For the first point, I realized that in previous updates I was going off of some assumptions from the datasheet that if we were above the gate-source threshold voltage, the PFET would be sufficiently conducting for balancing current to go through it with a small on resistance. However, I now realize that this threshold voltage is only for a drain current of -250uA, which is much lower than we need for balancing (balancing will be at ~400mA max).

Thus, I want to re-evaluate whether we are going to be supplying a sufficient gate-source voltage to the PFET to have it be "conducting enough" for balancing current. I will define conducting enough as having a small on-resistance (or small drain-source voltage drop) when we are at the minimum gate-source voltage we would be during normal operation. For reference balancing itself is ~10 ohms, so we want to be much lower than this.

For reference, the on-resistance of the PFET at $V_{GS} = -2.5V$ is ~0.1 ohms.

In the past I used this image to justify that the gate source voltage we had was going to conduct enough current for balancing, however I now realize that I miss the critical equation at the top left which says this is only true at $V_{ds} = -5V$.

For our case, we can estimate that if at $\sim 0.1\text{ohms}$, the V_{ds} will be $(0.1\text{ ohms})/(0.1 + 10\text{ ohms}) * 2.5\text{V} = 0.025\text{ V}$ or around 20mV .

[Missing image: ../../images/image_2691345278.png]

This is confirmed by simulation where if we set a balancing PFET to conduct by connecting its gate to a lower voltage, the voltage drop will be at $\sim 30\text{mV}$.

[Missing image: ../../images/image_2691355473.png]

With that in mind, the datasheet is kind of unhelpful in determining the minimum gate source voltage we need to supply 400mA . We can try to use this chart from the datasheet below. Trying to determine what the drain current at $V_{DS} = 40\text{mV}$ will be a fruitless endeavor since that's like a very small percentage of the box on the bottom left. Instead, we can compare to something we definitely know works, we know that the PFET works fine when the V_{GS} is at $\sim 3\text{V}$ just based on the balancing PFET working when the modules are at 3V .

[Missing image: ../../images/image_2691365987.png]

Thus, if we work based on estimation, we see that the behavior of $V_{GS} = 2.5$ approximately matches $V_{GS} = 3.0\text{V}$, and so if we can get the V_{GS} to around this range, it will likely work. In other words, since the different V_{GS} curves become approximately the same at the V_{DS}/I_d range we are thinking about (30mV , 400mA), as long as the V_{GS} is approximately 2V or greater, the resistance should be fine.

However, instead of just speculating and saying it should be fine, we will test this with the autoconnection circuitry breadboard. However, this gives me more confidence than before.

Based on the above, I realized I should change the resistance value of the pull-up resistors to make the V_{GS} value bigger. This is only really an issue when the modules are at small voltage ranges (2.5V - 3V), but essentially, we want the the voltage value to be higher. If we use 10Meg ohm pull-ups, we get $V_{GS} = -1.4\text{V}$ at minimum.

If we instead use 22 Meg ohms , we get -1.8V V_{GS} , with the only downside being having to wait $2\times$ as long (~ 5 seconds).

Note for reference, we can also use 33 Mega Ohms to get at V_{GS} of -2.0V , and so we may switch to using that if necessary. For now, I think I will switch to 22 Mega Ohms for a balance between speed and conduction.

We can create this circuitry pretty easily by changing around our old circuitry to roughly match the circuitry they use.

For the Gate-Source voltage curves, we get something that looks like this. We see that each gate-source voltage varies ALOT all the way from 2V to 20V .

Zooming in on the beginning (when connection occurs), we see this interesting effect where the highest voltage modules have the steepest time constants, before they settle at steady state.

Finally, we can look at the capacitor voltages at the input to the ASIC. We see that here the voltage curves are a lot more messy (i think), and that the order of the voltages being connected seems to be a little over the place. Specifically, I found that some adjacent modules seem to get connected very close to next to each other, and we see a lot of noise.

[Missing image: ../images/image_2691466878.png]

If we can get anything from this it is that despite Formula E's circuitry being much less "stable" as far as I can tell/simulate, we know that it still works and doesn't fry the LTC6813 based on the fact that they brought up the board and it worked. Therefore, I think we should prioritize having a low on-resistance and low-quiescent current for the FET rather than making sure that the connection and disconnection curve have a sufficient amount of time between modules. (In the past we said we wanted at least ~1ms between modules being connected, but I no longer think this is necessary). As long as the voltage curves don't spike above 8V or below -0.3V, we are fine.

Untitled

Author: Hemat Wander

Date: Jan 18

PCB Routing Update #2: After an afternoon of work, I have completed the routing for the temperature traces, along with connected the modules on both sides of the board (connecting module 8 to module 9 for the balancing and autoconnection circuitry).

Everything seems to be routed, however if its routed well is starting to seem a little questionable to me. The mess of routing you see below was my best attempt to get everything to fit on the top or bottom routing layers. My justification for this being fine is that the only fast changing traces will be the MUX select traces (given by the bottom right 2 pins on each IC respectively). As these traces are routed completely on one layer, and will have a full ground plane underneath them, I think it will be fine. However, these traces are decently close to the other temperature traces so there could be some cross talk.

[Missing image: ../images/image_2691182825.png]

- Another concern is that the temperature lines cross over the area of the buck converter which will have a lot of changing voltage due to the buck converter, which might couple to these lines producing some noise. As before, the only reason I routed like this was because of routing constraints. If we want to make it not enter this area, we can instead route the temperature line on the 3rd layer instead of the bottom layer (as was done on V4). I'm still considering doing this, however it may not be necessary. Doing so would cut through the ground planes, however it could be an option. @Krish D What are your thoughts?

[Missing image: ../images/image_2691184559.png]

- Finally, there is the balancing trace I had to route essentially all the way across the board. Because the resistance of balancing is only 10 ohms, the resistance of each trace has a much greater percentage effect on the resistance of the line. In this case, by doubling the width of the trace, I got the resistance down to ~0.15 ohms, which I think is sufficiently small (1% difference in the resistance of the line).

[Missing image: ../images/image_2691186893.png]

- I have a few next step considerations for routing. Mainly, how am I going to route VREG, VREF2, and ground fused. In V3 this was done using the 3rd layer, however if we want to keep that as a standard GND layer for stability, then we may want to route this using a polygon pour on the bottom and top layers. The main issue is that the fuses for VREF2 and GND are at the top right of the board (since they fit there), while the the sources of VREF2 and VREG respectively are in the center of the board. I'm a little unsure of how to route this.

[Missing image: ../../images/image_2691191453.png]

- Final consideration: Based on the last point, I'm considering if it is better to get rid of the fusing for the GND of the temperature lines as we already have a GND fuse from the GND connecting to the board. The purpose of this fuse originally was to isolate shorts on the temperature circuitry from the rest of the board, however I'm considering now if this is still necessary. The issue as I explained above is that the fuse is so far from the source, so all the modules on the bottom of the board will need to have traces traveling all the way up and then down again.

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Krish D (Jan 22)

@Hemat Wander Great progress so far!

- I don't think it is a good idea to have any traces cut into the GND plane nor to have them run close to inductor and buck converter for the same reasons you mentioned of noise isolation. Given that the two GND layers have a larger distance between them, and we are also not expecting the dv/dt of the temperature lines to change quickly, is it not feasible to run them through the GND plane to reach the MUX?

- For the routing of VREG and VREF2, is it not feasible to add the fuse closer to the center of the board near the MCU? It seems less risky to me to have it closer to the MCU. I'm also a bit unsure where the polygon needs to be routed. Can you make this more clear?

Given that so many power polygons would be needed to move through the ground plane, I'd be a bit concerned about routing any of the sense lines over the traces (Of course we don't expect to route high current to travel through these traces, however I'm not sure what transients may exist in the power lines during startup or when the capacitors are powered). This being said, what are your thoughts on making the slaveboard **6 layers**?

CC: @Aarjav Jain

Untitled

Author: Hemat Wander

Date: Jan 15

PCB Routing: Since we agree that the connector placements are good to go we can continue with routing. As a summary of what we expect the module wiring to be, here are the module numbers

[Missing image: ../../images/image_2687352651.png]

Schematic Changes:

I realized there was no resistor in series with the opt isolator, which means it would have shorted to ground if connected to VREG. To prevent this I added a resistor to make the current 10mA for the SPDT relay as recommended [in this datasheet](#). and 4mA for the other relay as recommended [in this datasheet](#). To get these currents, we assume a ~1.4V drop across the diode as explained in the datasheet, and a ~0V drop across the channel of the NFET.

$R1 = (5-1.4)/4\text{mA} = 900 \text{ ohms} \rightarrow \text{take } 1\text{k ohms}$

$R2 = (5-1.45)/8\text{mA} = 450 \text{ ohms} \rightarrow \text{take } 470 \text{ ohms}$

[Missing image: ../../images/image_2687266137.png]

- I changed the input to the buck converter to be coming from the in pack module regardless of if we are in scrutineering mode or not. Previously, this input came from the output of the SPDT solid state relay used for scrutineering, as that would have changed what the max voltage was. However, for the buck converter, we don't need to care about using the external pack module. Furthermore, going through the solid state relay would add resistance on the line going to the buck converter.

[Missing image: ../../images/image_2687269655.png]

Routing:

For the voltage lines coming from the connectors, we want to use a width that can at least take 400mA. For our case we will rate the width for ~four times this value, as we have enough space to do so. Thus, we can use 0.635mm width, to get a max current rating of 1.43A. The purpose of doing this is to reduce the resistance of the line, as we have a 10 ohm resistor for balancing, and thus changes by 0.1 ohm or such will more drastically change how much current flows (as compared to most of the other traces)

I also used similar width lines for all of the buck converter circuitry, to ensure that all of the high current draw elements of that circuit has a sufficiently rated trace.

I have now completed routing for the analog circuitry, internals of the buck, scrutineering and temperatures outputs going to the resistor divider. The bulk of the remaining routing is isospi and the temperature lines. After that I have to figure out where the polygon pours will go.

Aarjav Jain (Jan 25)

@Hemat Wander: What is the resistance of the trace if you make it rated for 400mA? Or I guess why is the resistance too large? What problem is it causing?

Untitled

Author: Hemat Wander

Date: Jan 8

PCB Layout Update #3 I've completed another draft of the [slave board here](#). The dimensions of the slave board are now 120mmx160mm tall, because I realized that I could increase the height of the slave board while still ensure the balancing LEDs stay within the viewable window. Thus, the components are now much less cramped, however the size of the board had to increase as a result. The two downsides to increasing the size are 1) the cost will be higher for the PCB, and 2) the wires will have to reach further across the board to reach modules on the other side.

****Note:** The overlay "boxes" were mostly made for myself in understanding where the components for each schematic section were, but we can remove them if needed.

****Viewing Balancing LEDs:** The polycarbonate for viewing balancing LEDs will be 70mmx200mm which we now fit within, due to moving the balancing LEDs to the center. This distance from one side to the other end of the balancing LEDs is <85mm, meaning that we could have the two slave board next to each other, with a ~30mm space between the PCBs (based on the 3D printed holder) and still fit within the window. @Deev Shah @Krish D @Samuel Shin

[Missing image: ../images/image_2669487027.png]

Small schematic changes:

- I removed these test points since I don't think we need them (@Krish D if you have any thoughts)

[Missing image: ../images/image_2669464320.png]

- I changed the Buck converter's inductor back to the previous revisions 220uH as opposed to the new 1mH despite us calculating 1mH based on the datasheet. The two reasons for this are because we want to reduce flux leakage leading to EMI on this analog board, and the second reason is because we already know that having 220uH already works from the previous slave board. I'm not entirely convinced of the first point because some sources say shielded inductors have virtually no EMI, but to the second point if we frame the inductor choice as needing to justify a **change** from the previous revision, then I can't really do that as we didn't have any reason that changing the inductor was needed. - I also realized that I was missing a capacitor at the input, which I have now added.

[Missing image: ../images/image_2669468342.png]

- Thus, the only changes from the previous revision buck converter circuit, is adding a 5.5V breakdown TVS diode at the output, and adding a 0.1 ohm resistor in series to the output capacitor to improve stability.

Changes to the board layout: - As mentioned the main change was a change of dimensions. This allows us to spread the components more vertically, creating more space between the SMD components. This also allows us to create more space between the isoSPI circuitry and the measurement.

[Missing image: ../images/image_2669472380.png]

- I removed the temperature resistors and capacitors from between the plastic connectors, as I realized they would be very difficult to reach afterwards with a soldering iron or heat gun.

[Missing image: ../../images/image_2669473162.png]

I labeled and added all the test points, meaning all of the components are now on the board (yay!)

I changed the scrutineering connector to a 4-position to match the module board.

[Missing image: ../../images/image_2669475859.png]

Concerns:

One of my concerns is that the voltage input fuses will be difficult to desolder and resolder in practice due to the placement close to other components (including the test points). We likely won't have to do this often (based on previous experience). If we did want to improve the fuse positioning, I'm not entirely sure how we could do this, so if you guys have some thoughts that would be helpful @Krish D.

[Missing image: ../../images/image_2669477296.png]

Other than that I don't have many concerns, as the ASIC now has more room in the middle. The only thing is if we want to decrease the dimensions for cost savings.

[Missing image: ../../images/image_2669479509.png]

Room for improvement?

- There is now some empty space around the board, which is good for keeping the analog measurements at some distance from the isoSPI, however it means that we could shrink down the board dimensions if we wanted to, although its not entirely clear how this would be done in practice.

Krish D (Jan 9)

@Deev Shah @Samuel Shin Can you folks please comment on whether or not the size poses any feasibility changes for scrutineering, module access, etc.?

Samuel Shin (Jan 9)

@Krish D Not sure which size you are talking about. If you are talking about the 30 mm gap, I think you guys have much better understanding on how it could improve accessibility for scrutineering.

@Hemat Wander One question I had after reading the update. Why does increasing the size of the board make wires need to reach further away? Shouldn't it be decreasing the length of the wire?

Hemat Wander (Jan 23)

@Krish D

I found some documents that help give some information on component selection. 1. [Selecting Inductors](#) 2. [General selecting components](#)

As from the calculations I found before, it seems that a higher inductance value reduces the current ripple.

[This random person on the internet](#) gives an explanation for why HIGHER inductance values are better on LOWER loads. He explains that for low-consumption loads, sometimes the output capacitor doesn't get discharged fast enough to justify the buck converter charging power into the output capacitor. This means that the converter will instead "skip a step" essentially meaning it will not charge up the output capacitor in this time. This can lead to stability.

In our case, I added a 0.1 ohm series resistor and output Zener diode to help with stability and quick large increases in the output voltage, however increasing our inductance value might further help with stability. If we want to improve stability, a higher inductance value seems better.

HOWEVER, the buck [converters datasheet's](#) application notes has a recommended inductance value for a 5V output specification (the 220uH we used last time).

This is essentially now a question between, do we want to stray from what we know works and what the application notes suggest for the chance of increasing stability. If the answer is yes, we can change to a 1 mH inductor (or something in-between), otherwise we can keep the 220uH inductor.

@Krish D

Krish D (Jan 24)

@Hemat Wander

From my understanding, having a lower load implies that it will have a higher current draw and requires current stability (i.e a lower output ripple at the same). Would we consider the V4 revision of the slaveboards to have a high enough current draw to require this stability?

It's hard to know this without testing since the buck converter datasheet has it's own recommended value, which is why I feel it makes more sense to go with the 220uH inductor since it was what we used last time (As we should follow what the manufacturers suggest for ideal performance).

To finalized this decision and to try and be practical, is it feasible to find higher value inductors and lower value one that was used on the previous revision of the slaveboard with the same if not similar size footprints so that the noise level can be tested?

Hemat Wander (Jan 24)

@Krish D Lower load in this case means higher resistance not lower resistance (AKA lower current draw). I agree with that and will look into how similar the footprints will be between 220 uH and 1 mH tomorrow.

Author: Hemat Wander

Date: Jan 3

PCB Layout Update #2 I've completed an initial draft of the circuitry for the [slaveboard here](#). For this draft I'm just trying to see generally where everything will be placed and then I will finalize everything and fix any small mistakes afterwards. The main thing I'm thinking about is if the number of components we have means we will have to expand the board size to greater than 120x120mm for its dimensions. Currently everything seems to more or less "fit" give or take some of the test points. However I'm unsure if having the components this close together is a good idea (I discuss this below).

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Possible Optimizations:

Can move some of the autoconnection or balancing/filtering circuitry to the space between the connectors, as it seems like there will be - Possibly could optimize the placement of the autoconnection circuitry + balancing/filtering SMD components to have them be more cleanly packed, and thereby save some space.

Concerns: - All the SMD components being so close will make bring up more difficult and also increase the risk of neighboring components shorting together due to unnoticed solder or wear and tear.

- The isoSPI line will be too close to the analog measurements, thereby leading to noise in those voltage readings. Currently I'm thinking I will surround the voltage measurement (C1, C2, etc.) traces with a pour of the previous module voltage to create a stable "return path" and surround the isoSPI with a neutral pour "chassis" that isn't connected to the main circuitry in anyway.

- The ADMBS1818 will be too close to the passives. This is just an issue for bringing up the board and debugging if we have to replace the ADBMS1818 it will be more difficult.

There is not enough space for labeling resistance values. Specifically, I wanted to label the resistance values of the resistors in the autoconnection circuitry that can change value, however I doubt there will be enough space as is to do so.

The inductor is too close to the analog circuitry. Same idea as the isoSPI being too close, except more so because inductors have a lot more flux. Based on how close the inductor is (~7mm) to the nearest voltage measurement capacitor, I'm currently leaning towards reverting back to a lower inductance value (as suggested by @Aarjav Jain) so we can have less flux and thereby reduce the chances of voltage measurement noise. Furthermore, we want to avoid the catastrophic issue of isoSPI communication being messed up, as that is a higher priority than reducing current spikes from the buck converter.

The balancing LEDs might be outside the 200mmx70mm dimensions we have set. Currently they are in the correct height range (They only take ~70mm), however they seem like they will require ~240mm or more if we want to be able to see the diodes at the edge. This is because with the circuitry we have, it seems that the diodes will span the width of the board which is 120mm across. Thus my next priority will be trying to figure out a way to get it to fit within this range. CC: @Krish D @Deev Shah @Samuel Shin

- Also I have a possible concern for the schematic: will the TVS diode interfere with the algorithm of the buck converter as it serves as another diode pathway (although it is past the inductor as opposed to L5.1). Ideally it shouldn't because VREG should stay ~5V with some ripple, however this was just a thought I had I thought I should write down.

[Missing image: ../../images/image_2653623525.png]

Change to Scrutineering circuitry:

I realized we can use one FET for both SSDs as they can easily take enough current for both SSDs and we will thereby reduce current draw.

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Two options for SMD components: Note: The first 8 modules are all connected on one side (top), and the second 8 (9-16) are all connected on the other side (bottom).

For the first 8 modules I made all the components have the same sort of layout that looks like the following. This was just based on trying to optimize the component placements for having all neighboring nets next to each other.

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For the 9-16 modules, I used a different layout for the circuitry because I thought it would have to be different due to the FETs not being symmetrical when mirrored. I then later realized that I could have just rotated the components 180 degrees. Before I realized this I already created a separate layout for the bottom that has a slightly different organization of the components, and so we can now compare them and choose one or possibly keep both.

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Next Steps:

I want to decide if we want to keep the SMD components as they are or change how they are placed. That being either we expand the size of the boards to have them further apart, or we place the SMD components on the underside. - I also want to try optimizing the balancing LED placements so I can get them within a 200mmx70mm dimension rectangle. - After that I will work on labeling each of the modules so it clear what module connects to what. For reference, everything is kind of placed like this.

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Aarjav Jain (Jan 5)

@Hemat Wander

The isoSPI line will be too close to the analog measurements, thereby leading to noise in those voltage readings. Currently I'm thinking I will surround the voltage measurement (C1, C2, etc.) traces with a pour of the previous module voltage to create a stable "return path" and surround the isoSPI with a neutral pour "chassis" that isn't connected to the main circuitry in anyway.

Also I have a possible concern for the schematic: will the TVS diode interfere with the algorithm of the buck converter as it serves as another diode pathway (although it is past the inductor as opposed to L5.1). Ideally it shouldn't because VREG should stay ~5V with some ripple, however this was just a thought I had I thought I should write down.

Hemat Wander (Jan 6)

@Aarjav Jain

Your explanation of the neutral pour becoming an antenna makes sense, what I was thinking was to ensure that any return path coupling would occur on this neutral pour as opposed to one of the analog lines. Would the best way to do this be to fill everything with GND? What is FR4? @Aarjav Jain

- Sure I can investigate using 0603, however I will use that as a space optimization if required, as using 0805 components seems much more preferable for bring up.

120mmx120mm was a requirement we made based on the balancing LEDs needing to fit within the 70mmx200mm polycarbonate sheet on the control board, so 70x200 for both boards balancing LEDs is the absolute requirement. However I'm still figuring out how to best meet this requirement.

Currently the TVS diode is rated for beginning breakdown at 5.5V and clamping at 14V. So the the leakage current at 5V-5.5V is max 10nA, so I think we should be fine in terms of leakage power.

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- The order of module connectors can entirely be based on the position of modules in the pack. ([See this update](#)). However, I found that regardless of which order we place the module connectors, we have issues of wires going to modules on the other side of the board, and so it makes more sense to prioritize sensical ordering of the modules for clean routing and debugging, as opposed to for reducing wiring length. Does that make sense? @Aarjav Jain

Aarjav Jain (Jan 12)

@Hemat Wander yes makes sense. I had seen your update as well!

Untitled

Author: Hemat Wander

Date: Jan 1

Begining PCB Layout: I had to do some work cleaning up the footprints for each of the components.

- For the [SN74LV4052AQDYRQ1](#), I had to use the footprint + 3D model from [SN3257QDYRQ1](#), as they have the exact same package (16-SOT-23 THIN) and the former did not have a 3D model/footprint available.

- For the connector positions, I decided to have all of the first 8 modules on one side and the last 8 on another. This is just to simplify routing for the autoconnection and balancing circuitry. I can justify doing this as it is already quite impossible to have perfectly clean wiring ([as discussed in this update](#)).

- I tried creating a layout with the autoconnection components that would allow us to work without switching layers, however I got to a point where doing so was no longer geometrically feasible. My concern with the autoconnection circuitry, is that if we switch layers, there will be some added inductance or capacitance of the lines that will stop the circuitry from functioning as intended.

However as the capacitance values are 10nF or greater, and the resistances are only up to 10Megohms, the effects of routing changes should theoretically be insignificant. The risetimes are on the order of milliseconds, and therefore the frequencies should be fine. Thereby, the priority should be ensuring that the analog signals (the voltage measurements) are as straight and short as possible.

Also need to keep in mind we want to label the resistor values for the resistors in series with the autoconnection circuitry diodes. As those resistor values will change over time.

I'm beginning to get concerned if it will be possible to fit all the SMD components on one side of the board. Everything right now is looking very tight. If we moved all the SMD components to the other side of the board, and only kept the LEDs and ICs on the top, it might work better. However, I will only look into doing that if necessary.

Also possible concern, will balancing current exceed the limits of the FET for the VGS we are operating at? [It says that](#) at 2.5V VGS voltage the max current is 2.5A, so we might have to look into if at our min cell voltage, we are still in a good current threshold.

Aarjav Jain (Jan 5)

@Hemat Wander Pack will fault at > 2.5V. Does this change the concern mentioned at the bottom?

"I'm beginning to get concerned if it will be possible to fit all the SMD components on one side of the board" -> As you found out, correct!

What do you mean by change the resistor values? And why will they change over time?

If 4 layers is genuinely not doable with our board size requirement we may consider a 6 layer PCB.

Hemat Wander (Jan 6)

1. Ideally the current through the PFET will be fine, however it is kind of hard to tell from the datasheet, if the on-resistance of the PFETs will be significant at lower cell voltages. The concern here is that the on resistance is significant enough to affect balancing current noticeably or even the voltage readings. Since we are ideally only going to be balancing at higher voltages, it should be fine, but I haven noted this down as something to test with the in person testing circuitry.

<https://www.diodes.com/assets/Datasheets/DMP2110U.pdf>

[Missing image: ../../images/image_2661419786.png]

2. The resistor values will change between each autoconnection **module**, and they vary gradually between 100k ohms and 4.32M ohms
3. The layering of the PCB is not an issue, there is ample space for routing I think, the issue is fitting all SMD components on one side of the board.

Untitled

Author: Hemat Wander

Date: Dec 2025

Connector Placement on Slave board: Due to the non-mirrored layout of the modules within the pack, it isn't exactly obvious how we should place the moduleboard-slaveboard connectors. There are a few different options we have, and none seems particularly clean or ideal. We had a [brainstorming session previously](#), where we came up with some ideas of how to place the connectors. The three types of board layouts we came up with is either

1. 4 connectors on the top and bottom

[Missing image: ../../images/image_2648185868.png]

2. Connectors all towards the edge of the board, with the second board then having to be rotated.

[Missing image: ../../images/image_2648187106.png]

3. A skinny PCB, with all of the connectors on one side.

[Missing image: ../../images/image_2648188138.png]

If we went with option #1, because there is no symmetry between the first 16 and last 16 modules, the wiring would have to be a little messy. That means that we would have some wires crossing underneath the slave board, and over from one side to another. I came up with one possible option of placing the connectors below, however other options would work as regardless of what we choose, it is going to be messy.

The main benefit of this option is that the slave boards will be facing the same way, meaning the balancing LEDs and other things we need to check will also be facing the same way. I.e. its less confusing.

[Missing image: ../../images/image_2648190997.png]

In the above drawing, the numbers represent the module positions, and the 1/2 represents that connector connecting to module 1 and module 2.

Note: We could further make this option "simpler" by having the first 8 modules connect to one side of the PCB or on the other side. Doing this however, would make the wiring more slightly messy and increase the overall average length

of wiring.

If we went with option #2 we would be prioritizing lower overall wiring length and slightly cleaner wiring. The main downside is that the PCBs have to be rotated from one another, meaning it might be slightly more confusing to look out (figure out where balancing LEDs are for), etc. The other downside is that it's not exactly clear where we should plug in each connector to the module boards as they are extremely out of order. This just means we would need to label very well or used coloured connectors.

Note that regardless of if we choose option #1 or option #2, we would have wires crossing over each other and under the slave board. However this option will have slightly less so.

[Missing image: ../../images/image_2648195550.png]

Finally, if we chose option #3, I think we would simply have all the connectors placed in order from module 1/2, module 3/4, etc. down the length of the board. Again, the second PCB would have to be rotated 180 degrees from the first one, meaning it would again be confusing when we are trying to figure out where to probe/which balancing LED goes to what, etc.

Also there might be more wasted space on a skinny PCB in terms of placing components, however I'm not entirely sure about this.

I currently think we should just go for option #1, with perhaps a change in the ordering of the modules for each connector (but still having 4 connectors on top and 4 on the bottom). This would sacrifice wire cleanliness, but make the board simpler to connect to the battery (as the connectors will connect to the modules in order).

CC:

Krish D (Dec 2025)

There are two ideas that come to mind here:

1. Is there a reason we should place heavy emphasis on the wires being in a mess?

- Of course elegance is partly the goal here, but practically, we won't need to be accessing the module boards often, since there will only be passive components and connectors. Therefore regardless of if the wires are arranged well or not, we will have to interact with them the same amount in both cases.

- Increased length means a higher likelihood of snagging cables, however this can be mitigated by better wire management (zip-tie-ing wires for example to the modules or in labelled bundles). Also, the same problem was present in the V3 pack when it came to the length of the harnesses having enough slack to get caught on things).

2. Convenience of debuggability:

- Having longer wires at the expense of ensuring LEDs are aligned intuitively for debugging seems like a no-brainer to me. Since it is unlikely that the connectors will have to be taken off multiple times (since we will have the SBTPCB to perform tests), the priority should be visibility.

For these reasons I agree that connector layout option 1 makes more sense to go with.

Untitled

Author: Hemat Wander

Date: Dec 2025

Slaveboard Schematic Cleanup:

I added bypass headers for the autoconnection circuitry, so that we can always use these if the autoconnection circuitry ends up not working. These will work in the same way as they did on the previous slave board. However, these headers can also serve as test points, meaning we can remove the test points we already had for those two nets at each module.

[Missing image: ../../images/image_2647935651.png]

[Missing image: ../../images/image_2647941989.png]

- For the Temperature circuitry, I only included test points for the MUX selecting and MUX outputs. This is because I expect us to be able to probe each individual temperature using the SBT (slaveboard testing PCB).

[Missing image: ../../images/image_2647942490.png]

- I also found an explanation for why we skip C_12 in the datasheet of the ADBMS1818 when measuring 16 cells. It explains on [page 86. of this datasheet](#) that we can either have all the unpopulated cells be at the end of the last MUX (skip C_17 and C_18), or we can choose to skip C_12 and C_18. Skipping C_12 allows us to "optimize measurement synchronization" by equally distributing the skipped cells among ADC2 and ADC3. I'm not entirely sure what this means, but since it was effective last time, and I don't think it is that messy to skip C_12, I will keep it as it is. Just note that we could skip C_17 if we wanted to. CC: @Aarjav Jain @Krish D

[Missing image: ../../images/image_2647950836.png]

[Missing image: ../../images/image_2647992690.png]

- I also added numbers for all of the components based on the schematic page (R2.5, etc.)

Next Steps: - Auto-connection testing BOM/procedure - Connector placement on the slaveboard

Untitled

Author: Hemat Wander

Date: Dec 2025

Testing Auto-Connection Circuitry Plan:

During the slave board schematic check-in we decided it would be a good idea to perform a real-life test for the autoconnection circuitry, to see if it is likely to be effective. The circuitry we will test with looks something like the schematic below, where we will have 6 modules each attached to a capacitor and resistor in parallel to simulate the internals of the ASIC draining power. The overall goal is to be able to see a >1ms time difference in the voltage charge-up between each module.

How will we construct this: As mentioned, we will use a solderable breadboard with all of the passive components seen (resistors, capacitors, diodes and PFETs) being SMD components. For the cells, we will use power supply's to effectively act as separate voltage sources. For this reason, we would need at least 6 separate power supply outputs for this to work. If not, we would have to use a fewer number of modules (I put 6 because I remember us having 6). How many power supply's do we have @Aarjav Jain @Krish D ?

What is the point of this test?

As this circuitry is a little complex, and relies on very extreme values of resistances and currents (megaohms and microamps), along with the variable threshold voltage of the PFETs, we want to test that this circuitry behaves as we expect from the simulations.

In general, we will bring up this circuitry (all of the components seen) on a solderable breadboard, and then test using an oscilloscope to compare the voltage across each adjacent capacitor, to see if they charge up one after another. Specifically, we will perform at least 5 different tests, where we either compare module 1 and 2, module 2 and 3, module 3 and 4, etc. On top of that, we would also likely want to compare gate-source voltage waveforms for neighboring modules. Note that I'm saying we can only compare two voltages at a time, because the oscilloscope only has 2 input channels.

To make it crystal clear, we would be having each oscilloscope channel across one of the capacitors, that we would normally have connected right before the ADBMS1818 (the 10nF capacitor).

As I haven't really tested such a small time-scale transient before, we will have to figure out how to properly trigger and capture waveforms of this sort. I wrote [a doc](#) that may help with this.

In general there are only a few things that we could imagine being different between the simulations and real world behavior:

Thus, on top of getting waveforms for the voltage across the capacitors, we would be on the lookout for seeing in what range the values are off from expected simulations (if it works at all). Furthermore we would also want to test some of the extremes mentioned in the [previous update](#), so changing some of the power supply voltages, and seeing how the waveform behavior changes (time wise and voltage wise).

Change to Timelines: Doing these tests doesn't necessarily mean we have to delay the design process (since I can't test since I'm back in January), however we would have to delay ordering the PCBs by necessity, to have the tests inform if the circuitry works. Thus, when we come back on the week of Jan 5th, if we could ideally get started testing and complete by the 20th. Then, if everything works as expected, we can order the PCB now by the 20th. This is of course later than we were expecting before.

However regardless, the PCB should ideally be routed by the time we start doing all this testing, so that we are good to order if everything works. How would this sound @Krish D @Aarjav Jain

Krish D (Dec 2025)

@Hemat Wander

Technical questions/notes:

- We have a 6 power total power supplies that we can use. Keep in mind though that this would make the wiring of your setup quite tedious and a pain to deal with safely. Is using the resistor ladder divider not a feasible option?

- For the setup itself, how many points need to be probed at any given instance? Did you plan out where the test points would need to go in this case? A circuit diagram on the breadboard itself (similar to what @Christopher De Lazzari has made the pre-charge check circuitry would be ideal).

- I don't think we should have any issues with looking for the transient since it will be on the order of $<1\text{ms}$, so the procedure you outlined in the ISO-SPI test plan doc seems reasonable to work with.

- I'd like to see a more specific test plan and bring-up plan that outlines what points are going to be probed and in what order (since timing between capacitors charging is of great importance here).

Timeline:

- I don't see this impeding slaveboard-masterboard testing timelines, however I do believe the testing will not take longer than 3 work sessions (you've specified a total of 5 @Hemat Wander) Unless you for see some failure modes that prevent you from obtaining the $\sim 1\text{ms}$ gate-source voltage becoming $\leq -1.0\text{V}$ AND have a plan for how to remedy this (replacing specific resistor or capacitor values), than I can't imagine that probing and iteration will take much longer.

- @Aarjav Jain I'm ok pushing timelines for ordering the slaveboard by around 5 days if you don't for see any challenges that come with this.

Hemat Wander (Dec 2025)

@Krish D

Using a resistor divider ladder will absolutely not work, as the voltages will changes based on current draw (which doesn't emulate a the real circuit being cells in series) - I could imagine wanting to probe everywhere around the circuit to see voltage drops across the various components. However, if we want to have test points for places where we will be probing often, then I think we should have test points at the gate, source, and drains of the PFETs - Sounds good, I think it would be better to make the test plan after we order the components because we should order ASAP. - I'm putting 5 work sessions as a safety net, as in person testing usually always has some complications related to it.

Krish D (Dec 2025)

@Hemat Wander

Sounds good. Please add a small diagram showing where the test points should be placed then on a perf board for greater clarity.

Can you please make a BOM for the autoconnection circuit as your next steps? In the meantime, @Aarjav Jain please note I've changed the SB bring-up timeline to account for testing time.

Untitled

Author: Hemat Wander

Date: Dec 2025

Auto-connection circuitry Update #6

After a [check-in meeting on Sunday](#), we realized that we only performed simulations for the scrutineering circuitry under the condition that the cells were at 4V. Thus, we should test the simulation under different voltage conditions.

First I tested under the cells all being at a minimum voltage we can expect plus some safety threshold of 0.1V, giving us 2.4V. This is based on the cells having a minimum of 2.5V, so its likely we won't go under 2.6V. Under these conditions I found that the gate-source voltages of the PFETs were at -1.4V and below while the PFETs are closed (conducting). In comparison, I found that if the cells were all at 4.3V (0.1V safety threshold over maximum voltage), then the gate source voltages hover at around -2.7V and below. (Both charts are shown below).

As the max threshold voltage is only -1V (with -0.4V being the minimum), we find that even in the worst case for all cell voltages, we are still over the gate source threshold (so it will theoretically still work then).

Testing with a lower voltage unbalanced cells: Other than testing with all of the cells being at different voltages, we should also test for there being a reasonable imbalance between the cells and how that would affect this circuitry. Lets begin at an extreme to see the behavior. To begin, we can try testing module 5 being at a lower voltage than the rest of the cells (mod 5 = 2.5V, and mod 1-4,6-16 = 4.0V).

If we look at the gate-source voltage curves, then we see that the first 4 voltages behave as expected, but then every voltage after module 5 is much higher (less negative), meaning that the rest of the PFET gate-source voltages are pushed upwards because of the one faulty cell. Furthermore, we find that instead of the first module settling at the lowest (most negative) gate source voltage, and the rest of the modules being at slightly higher (less negative) gate-source voltages, we actually see the opposite pattern, due to module 5 being at a lower voltage.

If we look to the input capacitor voltages (to see the order they fill up in), we see that again, the first 4 modules fill up in order, but then after that it gets a bit sketchy. Namely, we see a long delay between module 4 and 5, and then after that, a lot of the modules seem to fill in right after one another, and then we finally see the last few modules filling up with a proper space between them.

If we zoom in, we find that modules 5-10 fill in a slight reverse order, meaning that module 10 fills in before module 9, and so on down to module 5.

I originally assumed this was bad because the modules are filling in out of order (meaning for example module 7 will be filled slightly before module 6). However, we need to back up for a second. From first principles, what we are looking for is that nothing breaks the max thresholds of the ADBMS1818 (in terms of max voltage between inputs or max current through the internal ESD diodes). In terms of the current, we find that the internal ESD diodes current stay around 22uA max, which is well below the threshold of 10mA. This is because the capacitors fill up so close to one another, that they are practically filling up at the same time.

For reference, if we plug module 5 in first without plugging in the previous modules first, we get a 81mA drop across the internal ESD diodes (way over the threshold). This would be an example of what it would look like for us to break the ADBMS1818 from a simulation perspective.

If we now test the opposite case were module 5 is overfilled to 4.1V, while the rest of the modules are at 2.7V, we get some more sketchy behavior. This time the gate source voltages of the first few modules are relatively low because of the 2.7V cells, however the 4.1V cell makes the 5th module gate-source voltage drop way down to -3.0V, meaning that all of the modules passed it overlap with the initial 4 modules.

I was expecting that this would mess the timing of everything up for the connecting stage, but surprisingly, everything seems to still connect in order. likely due to the behavior very early into the gate-source voltage curve. Furthermore, if we look at the current at the internal ESD diodes, we see that it says in the nA range (well below the threshold).

Finally, if we look at the disconnecting stage for this unbalanced case, we somehow miraculously see that the modules still disconnect in the correct order (last->first). I'm honestly not sure how this is working, but it does seem to be working.

We find that for the simulation, even under different voltage conditions for the cells (with the cells being in the 2.5V-4.1V range) or with there being an imbalance with some cell at a radically different voltage, we still find that even if the charge up behavior is a little sketchy, we find that the order of the connections is expected, or otherwise the current of the internal ESD diodes stays well within the thresholds. Thus, the circuitry still looks safe from this perspective.

Krish D (Dec 2025)

@Hemat Wander Amazing update! I'm glad you considered every edge case and defined from first principles if the discontinuities in proposed behavior is significant for causing damage to the ADBMS1818.

Can we finalize the design of the autoconnection circuitry as it stands and begin testing?

Hemat Wander (Dec 2025)

Yes, everything seems good for this being the finalized design for the autoconnection circuitry, and so we can test accordingly.

Untitled

Author: Hemat Wander

Date: Dec 2025

Auto-connection circuitry Update #5

Going on from [this update](#).

Context: To give some clarification of what the circuitry looks like, we have the following circuitry at each module (3 modules interconnected are shown). The PFET is what controls the switching from cell to the ASIC.

[Missing image: ../images/image_2635212753.png]

The top resistance is 100k for every module, but the resistance in series with the diode is different for each module. The first 8 modules all use 100k ohm resistors, but after that I found we have to steadily increase the resistance (220k, 470k, 680k, 1Meg, 1.5Meg, 2.2Meg, 3.3Meg, 4.32Meg). (@Krish D, adding these different resistance values is what I found to have "fixed" the simulation to get it working). I'm not entirely sure why these values work, but they seem to work based on the simulation.

SPICE simulations Now we can return to the spice simulations. The chart we see below compares the gate-source voltages for each of the 16 module PFETs. At $t=1$ seconds we connect the jumper and at $t=2$ seconds we disconnect the jumper. We expect that when we connect the jumper module 1 is the first to connect and module 16 is the last to connect. When we disconnect the jumper we expect that module 16 would be the first to disconnect and module 1 would be the last to disconnect. This does in fact seem to be the behavior we get. Note that we also get that the gate source voltages for the PFETs differ a bit between each module (-2.4V $\rightarrow$ -4V), however they are all under the gate-source threshold in the datasheet of -1V.

We can also plot the voltages across each of the input differential capacitors. This now directly shows the voltages being input to the ASIC across each of the capacitors directly attached to the LTC inputs. The first image shows them being connected, and we see that they turn on in order, but the time difference between voltage rises is anywhere from 0.01ms - 1ms. The second image shows them disconnecting, and we find that they disconnect in exactly the order we want (last-first). In other words, this means it technically works.

Increasing Capacitance As suggested by @Krish D and @Michael Lin, I tried increasing the capacitance value of the capacitors to increase the time between connections, and by switching from 10nF capacitors to 1uF, we get the following results. The connecting part looked very promising, with us getting at least 4ms between connections, even with the bunch of modules connecting one after another in the middle.

However, the disconnecting part is less great, as we are using 10Meg ohm resistors for pullups (to try and not disrupt the circuitry), we get that with 1uF capacitors, the time constant is much longer ($1\mu\text{F} * 10\text{Meg} = 10$ seconds), meaning we would have to wait at least 30 between disconnecting the jumper and unplugging the connectors, which might be tedious in a time-crunch scenario. What are your guys thoughts on this @Krish D @Michael Lin ?

The voltages here again represent the capacitor voltages input to the ASIC. We connect the jumper and see the voltages rise up fairly quickly, however if we disconnect the jumper (at $t=10$ seconds), we see it takes until $t=40$ seconds for everything to fully disconnect.

So in general we see that increasing the capacitance, doesn't majorly effect the connecting stage (as it is already in the ms range), however it will affect the disconnecting stage in that we will have to have an additional step in the procedure to wait longer before doing anything. Note that we can also have any other resistance value in the 10nF-1uF range. Calculating the time constant for the disconnecting would just be from $t = 10^7 \text{ Ohms} * \text{capacitance}$.

So a 100nF capacitor would have a 1 second disconnecting time constant. This currently seems like the best option so we only have to wait a few seconds (less chance for error). I will stick with using 100nF for now. This gives us at least 1ms between each connection occurring, and we only have to wait ~3seconds for the FETs to discharge when disconnected.

Something to note here is that we technically have no idea how long it will take for the capacitors connected to the ASIC (not the FET capacitors) to discharge, because we don't know the shunt resistance internal to the ASIC. However, this information shouldn't affect us, such as for our current slave boards, where once we disconnect the module jumpers, we technically don't know how long it takes for those capacitors to discharge.

Components used in simulation: Also I should specify the components used in simulation, and how well they match real components. For the diode I used a 1N4148 model in LTspice, and for the actual schematic I am using the 1N4148WX-TP, so it should be similar. For the PFET I'm using a default model with a 20V gate source max voltage and a 0.8V threshold voltage, so essentially something in the same range as our PFETs. I don't think its necessary to find an exact PFET model, as this design should work for a variety of PFETs.

Next Steps for auto-connection circuitry: We discussed in person if performing tests for this circuitry is a good idea before ordering the PCB. Originally I was not sure because of how much time it might take to perform these tests. However I think that performing these tests might be helpful if the behavior is just completely off of what we expect. We would have to make a mock breadboard to see if the voltages for the PFETs are within the correct range, and each adjacent group of modules are activated at least 1ms after each other. The other issue is that it would require 16 voltage inputs to plug into the PFETs, which is what the slaveboard testing PCB was originally for. For the last reason alone, I don't think we can perform the test. I considered performing a test with only a few modules and seeing if it works, but in simulation I found the timing is highly dependent on the number of modules used.

Currently it seems the most feasible option is to just order the board and see if it works, by first only soldering in the autoconnection stuff, and testing using the testing PCB. Then if it doesn't work, we have to order another PCB.

@Aarjav Jain @Krish D what are your thoughts on this?

Also, me and @Krish D discussed implementing FEs autoconnection circuitry in SPICE and testing it. Should we do that? I don't know what the solar standard for this scenario is, because it doesn't feel right to just use what they use? What are your thoughts on this @Aarjav Jain

Next Steps before schematic check-in: - Schematic cleanup, including net names and note colors, other standards, etc. - Modify buck converter circuitry including inductor and adding proper TVS diode - Check datasheets for integration points between components (ensure multiplexers are properly pulled up / down, and are analog not digital) - Add DNP common-mode choke capacitor.

Hemat Wander (Dec 2025)

Here is the LTspice file, for everyone else to check out. [Auto_connect_experimenting.asc](#)

Michael Lin (Dec 2025)

I think the 30s delay constraint might make operation more complex similar to how the "connecting in-order" constraint makes V3's board complex to operate. (If someone forgets to wait 30s before starting to disconnect the modules out-of-order, they'll cook the board, right?)

Maybe as a remedy to this, we can add one indicator LED at each module (or maybe just at the last module to disconnect?) so that we have light(s) that slowly fade away after disconnecting. This can indicate when it's safe to unplug stuff

Hemat Wander (Dec 2025)

@Michael Lin The issue with LEDs is that the circuit is going to be plugged in all the time, so we don't want to have a high quiescent draw source (LED) always lit up due to the circuitry.

Krish D (Dec 2025)

@Hemat Wander

Is it not feasible to test with 4-6 modules on modules on a breadboard?

I don't see the need to make all 16. We are looking for a general behavior since they are more or less functioning the same.

What are your thoughts though?

CC: @Aarjav Jain

Hemat Wander (Dec 2025)

@Krish D

In the past, when I tried simulating with 8 modules, I found that the timing behavior was very different, in that the first 4 modules had increasing time constants, while the last 4 modules had decreasing time constants. Essentially meaning that how close the module is to the beginning or end will decrease its time constant. Therefore, behavior with a different number of modules will behave very differently.

I can try simulating in SPICE what it would look like to use say 4-6 modules (what resistance values we would have to use), but just note it would be different.

Does that make sense?

Untitled

Author: Hemat Wander

Date: Dec 2025

Space taken up by LEDs on the slave board:

One of the requirements for the control board (@Samuel Shin) is deciding how much space will be taken up by the LEDs on the slave board, and therefore how much room on the control board needs to be reserved for polycarbonate.

Going from [this update chain](#), we know that the slave board will be approximately 120mm x 120 mm (max dimensions). This increase in size is both because we are increasing the number of passive components, and also because we are trying to ensure all the passive components are on one side of the PCB. This is mostly for ease of soldering, so we can reflow-solder all of the components on one side.

With that in mind, the circuitry for the the new slave board revision has a few more components being added in the chain of going from connector to ASIC. Approximately we have. The fuses and autoconnection circuitry is a new addition.

Connector -> Fuse -> Autoconnection Circuitry -> Balancing -> RC filter -> ASIC.

Note that the last 3 stages are the same as previously. This indicates that the space taken up by this circuitry will be approximately the same as previously. Although we will no longer have module jumpers, we will still have HV test points (for each module) and on-top of that we will have additional autoconnection circuitry between the connectors and balancing.

All of this is to say that the width taken up by the balancing LEDs and other resistors will be approximately the same as before. However, the height may differ based on how much additional space is required for the auto-connection FETs. Based on this, if we want to keep all of the components on one side of the PCB, we might need to further increase the width of the PCB to 130mm x 120 mm.

Regardless, the space maximum for the LEDs will be 100mmx100mm. This is to give space for whatever additional height we will need. Note that the width across is technically only 70mm between balancing LEDs, however I think that based on how the routing changes, additional width may be required.

[Missing image: ../../images/image_2602699713.png]

Hemat Wander (Dec 2025)

From the [last DR1 meeting for the control board](#), we decided that we are going to be having the polycarbonate panel be as small as possible. Thus, we are going to be using a width of 70mmx200mm. As mentioned, the minimum width for the slave board LEDs was ~70mm. And then we would have ~100mm per slave board in the window, meaning that the LEDs will likely be more centered toward one side of the board (as they are ~120mm each).

Untitled

Author: Hemat Wander

Date: Dec 2025

Auto-connection circuitry Implementation #4 Going on from the [previous updates](#), I realized that the resistance values we were using was much too low for our application, and so the quiescent current would be too high. I originally was thinking that I could increasing all of the resistances (say by a factor x100) and then call it a day, however, I'm weary about increasing the resistances so high all the way to 100 Megaohms, as there could be parasitic resistance effects

from the PCB itself + other components. I'm somewhat arbitrarily going to say that have 10Megaohms is a good maximum resistance value for us before we have to consider extremely strange parasitic behaviour.

With that in mind, I had to play around with the resistance values again. The goal was to get 100 kohm resistors between each module, which would get us to microamps of quiescent draw. After playing around with the resistor values for a while, I got to a point where all of the voltage values in the input to the ADBMS1818 are safe (don't spike to a high value), and the FETS.

I have attached the file here: [Auto_connect_circuitry_test.asc](#)

I have also changed the schematic accordingly.

| Aarjav Jain (Dec 2025)

| @Hemat Wander

Untitled

Author: Aarjav Jain - Electrical Director

Date: Nov 2025

@Michael Lin Great list to keep up with! Thank you for creating this!

@Hemat Wander Please be mindful of this doc as you finalize the schematic and begin layout/routing for the slaveboard. It would be useful to consider which tests need to be done and justify them to this doc, while also considering where you could add test points on the slaveboard to achieve them.

The update was originally posted on subitem Slaveboard-Masterboard Testing on 12-01-2025 at 02:04 by

Untitled

Author: Hemat Wander

Date: Nov 2025

Deciding on Max Dimensions of Slave board: We wanted to decide on a max dimension for both of the slave boards, and also the general layout within the slave boards. There are two options we have for the layout. Either the vertical option (seen below, which is what we have in our current pack) or the horizontal option (seen on the right).

In terms of determining the max slave boards dimensions, the connectors are the limiting factor. Lets assume we are using 4-pos connectors, as I think this will take more space than the other option of using 8-pos connectors, (thus

giving us a bigger MAX dimension).

Thus, if we assume that the space taken up by all of the connectors is $9.6\text{mm} + 2.5\text{mm}$ for width between, then we have $\sim 12\text{ mm}$ per connector. Assuming we have 8 connectors on two sides of the slave board, the width taken up by connectors would be $8 \times 1.2 = 9.6\text{ cm}$.

We then, would also have 1 extra centimeter on either side of the control board due to the mounting holes (assuming we use the same size). The total width is $9.6\text{cm} + 1\text{cm} + 1\text{cm} = 11.6\text{ cm}$ -> take 12 cm.

[Missing image: ../images/image_2586406608.png]

If we take a square area the approximate dimensions of the slave board would be $120\text{mm} \times 120\text{mm}$ which is compared to the original $100\text{mm} \times 100\text{mm}$. This is a 44% area increase.

However, we can just make the slave boards slightly rectangular to save area, as on the current slave board we still have enough space for the other connectors we would need (isoSPI, mock-scrutineering, etc.). Thus, we can take $120\text{mm} \times 100\text{ mm}$. (20% area increase from previous revision).

[Missing image: ../images/image_2586413775.png]

In terms of placement in the pack, having an horizontal orientation would be preferable for minimizing the distance between the modules and the slave board (decreasing wire length). The limiting factor for this is that the current iteration of the control board has more empty space in general towards the center, so doing something like below would be slightly problematic (this again depends on the exact dimensions, so we will confirm when we get those).

[Missing image: ../images/image_2586416417.png]

CC:

Krish D (Dec 2025)

@Hemat Wander Below is my thought process for assessing the feasibility of your chosen position for the slaveboards:

- I originally believed that it will be difficult routing wise to ensure that all LEDs can be kept in one block/grouped together. This will involve many vias and could become quite annoying to deal with, ***assuming you don't want to route across any of the power layers***. This is why I'd suggest keeping the LEDs where they are currently with regard to being next to each connector, and instead moving closer towards the center of the slaveboard (think a narrow column of LEDs that are on both sides of the ADBMS1818). However, this assumed that from [Sam's latest update](#), that you are choosing layout #2 referenced in my last reply.

[Missing image: ../images/image_2587747538.png]

- The other option is to route across the power layer, as currently on Brightside slaveboards, there are no traces on it! This would be an easy solution, I can believe this wouldn't contribute highly to EMI since the slave board has such low current draw, and having such a large GND plane will keep the continuous analog signals stable, as long as none of the balancing traces are being routed below the ADBMS1818.

[Missing image: ../images/image_2587747343.png]

Please let me know your thoughts

!

Aarjav Jain (Dec 2025)

@Hemat Wander

CC: @Krish D

Gurman Khella (Dec 2025)

@Aarjav Jain The chosen Minifit Sigma connectors are tin plated, so they are rated for a mating cycle of 30. The gold plated sigma connectors have a mating cycle of 250. For the gold plated connectors, I will have to find the specific combination of specific parts of the sigma series for this rating. Sometimes the datasheets for the gold-plated specific components also says 30, which is misleading with the initial datasheets, so I will have to confirm the specific components. Also, a note, gold-plated is more cost than tin-plated

Krish D (Dec 2025)

@Aarjav Jain I expect with our more recent changes (moving passive components to slaveboard), we shouldn't be messing with the module boards as often as we thought. The only times we needed to remove the connector from the cell boards on V3 was for pack disassembles and mod 32 for scrutineering, however this shouldn't exceed more than 30 times over the pack's expected 2 year life cycle.

Aarjav Jain (Dec 2025)

@Krish D That is true. Although I would like to see the cost and part number that @Gurman Khella is mentioning for higher mating cycles.

Untitled

Author: Aarjav Jain - Electrical Director

Date: Nov 2025

I have start a doc to list potential tests that we'd like to do. We can also add the actual test procedures to this doc in the future (under different document tabs)

https://docs.google.com/document/d/1R-2-kh3UOMei8B_aRGCagyDrYElhD4hJOsgBoEB7tsM/edit?tab=t.0

There are current 2 tests listed: measuring voltage amplitude & noise on the IsoSPI line, and testing the newly-added selfchecks

Next steps are to continue adding more tests to this doc as needed. We should also specify a date when we should finalize all tests to carry out (and write their procedures) before we finish bringing up the master and slave boards.

Untitled

Author: Hemat Wander

Date: Nov 2025

Scrutineering Test Circuitry: Going on from [this update](#):

The purpose of this update is to flesh out everything a little more by deciding on what connectors we will be using. Some initial notes: - We can use this component for the HV switches: [LCC110](#), but its quite expensive for the other switch, thus we can use a lower voltage switch - I also noticed when looking on Altium that SPST HV switches are much cheaper (\$2) compared to the SPDT HV switches (\$8).

Context:

I found this helpful link for comparing the types of relay options and their uses:

<https://www.ni.com/en/shop/electronic-test-instrumentation/switches/what-are-s>

Electromechanical relays are less expensive on average, (\$2) for SPDT compared to SSD which is (\$8) for SPDT. However, they are bulkier, meaning bigger size and more coil current required.

- Reed switches are used for high speed switching applications, and are generally less robust, which we don't really need.
- SSD relays seems to be the go-to choice for the HV switches we are going to be using, the only downside is that they are a little expensive (especially for SPDT compared to SPST).
- For the LV Tense switch we can also look at analog FET switching options, that essentially serve as Muxes.

Goals: - We want switches that are safe in that they provide high isolation resistance when not on. - Want them to be relatively cheap - Want them to not take up too much space on the slave board

Required Switches:

We need a HV SPDT switch for connecting the B+ of the C18 input to either the in-pack module, or to the external scrutineering module voltage - We need a LV SPDT swithc for connecting the T_sense line of the last module to the external module or in-pack module - We need a HV SPST (or SPDT if cheaper) for connecting the B+ of the 31st module (B- of the 32nd), to either be floating or to be charged. Purpose is for connection to scrutineering outside of the pack.

Some Options:

[J1031C5VDC.15S](#), this SPDT electromechanical relay is \$2 and rated for 60V, meaning it is a possible option for the Tsense line. The only downsides are that it is quite big (1.2cm x 0.7cm and 1cm tall), and that it draws a high amount of coil current (30mA)

[1462042-8](#), this SPDT electromechanical relay is \$7.11 and rated for a 200V load. It would be a good option for the B+ SPDT switch if we want a high off insulation-resistance solution.

[EE2-5NU](#), DPDT electromechanical relay, costs \$3.06 and takes 28mA of supply current. Could be used for both the B+ and B- connections.

[EE2-5SNU](#) DPDT relay similar to previous one, but is latching relay (meaning doesn't require continuous current draw, and costs \$3.62

[LCC110](#), high voltage SPDT switch, costs \$7.40. Only draws 2mA of control current. Could technically be used for all 3 switches, but will not due to the price. Because it is SPDT, we can connect both the normally open and normally close pin for our purposes, without needing to rely on using more switching circuitry.

[SN74LVC1G3157DCKR](#), 2:1 MUX (SPDT) which we should use for the T_sense line. Costs \$0.43, so I think it is the best option for this LV application.

[CPC1008NTR](#)(SPST- NO), a single throw SSD switch that costs \$2.49 and has 2mA of supply current. Could be used by itself for the B- connection. If being used for B+ connection, needs to be used in conjunction with another relay since it is SPST. Has 8 ohms of resistance (note that this is quite high in reference to the 10 ohm balancing resistance).

[CPC1125NTR](#) (SPST-NC), a single throw SSD switch. Costs (\$2.45), and has 2mA supply current. Could be used in conjunction with the above switch for the B+ connection to reduce cost.

So what should we use?: For the T_sense line, we should definitely use the [SN74LVC1G3157DCKR](#) mux since its cheap and small.

For the B+ and B- lines we have a few options.

If we prefer a more compact but cheap solution, we can use the [CPC1008NTR](#) SSD relay for the B- connection and [LCC110](#) for the B+ connection. However, this will be a little expensive (\$9.50), and will have some additional contact resistance (15-50 ohms on each side) meaning balancing will be off, at least for the B+ connection. These would only draw 2mA, and that too only during scrutineering.

If we want a bulkier solution, we can use electromechanical relays. Either with the [EE2-5NU](#) non-latching relay, or with the [EE2-5SNU](#) latching relay. Both of these would cost (\$3-\$4) **for both** switches together, but would draw 20-28mA at peak (during scrutineering), and take up more space on the slave board. One benefit is that the switching resistance would be smaller than in the SSD case. An issue with this is that it will generate some EMI during switching, however this should only be an issue during scrutineering?

****Important note.** These currents (2mA-20mA) would be drawn directly from the VREG line through a mosfet that we would control from a GPIO. The GPIOs themselves cannot support that much current draw. ******

CC: @Krish D @Michael Lin **What are your thoughts?

Hemat Wander (Nov 2025)

I was initially going to go with the most compact solution of the Solid State Relays. However, for the case of the B+ connection, the solid state relay would introduce too much resistance in the circuit for the normally closed (NC) connection (35 ohms), which is way too high considering balancing is 10 ohms.

Thus, for now I'm going to implement the circuit with the DPDT electromechanical relay. However, we might want to reconsider this in the future, given the higher EMI.

Hemat Wander (Nov 2025)

After discussing with @Krish D , we realized that we can just put the solid state relay after the balancing setup, as scrutineering doesn't have to worry about balancing. Then, if we set the RCfilter to be 68 ohms, the accuracy won't be harmed by much.

Thus, for now, we can stick with using solid state relays for both the B- and B+ connections.

Michael Lin (Nov 2025)

After discussing with these guys, we'll email the scrutineers again to confirm this. We're pretty certain this will be an okay implementation, but it wouldn't hurt to ask.

See original email chain:

<https://mail.google.com/mail/u/1/?ogbl#search/fwd+scrutineer/FMfcgzQcpnJldNxDeGrHMcFCkDpQHFTk>

Michael Lin (Nov 2025)

The draft email is now saved in battery email drafts. Please have a look @Hemat Wander @Krish D @Aarjav Jain !

Krish D (Nov 2025)

Hey folks, I just made a large lapse in judgement since I forgot we have 2 other cell inputs (ADBMS1818 has 18 cell inputs, not 16). Please take a look at [this](#) update for a revised implementation strategy.

Untitled

Author: Michael Lin

Date: Nov 2025

Component selection for IsoSPI connectors and Mux

Note that these components are not added to BOM. I will do so after finalizing these choices

IsoSPI line

PCB Header: [Digikey page](#)

Connector receptacle: [Digikey page](#)

Female crimp: [Digikey page](#)

Multiplexer: [TI's SN74LV4052AQDYRQ1](#)

Related changes

The required firmware changes to implement the new mux config should be pretty straight forward:

Hemat Wander (Nov 2025)

@Michael Lin Looks good for the multiplexer.

For the iso-SPI connector, I think we should use a vertical header as it will be going up through the control board. However this entirely depends on the pack geometry @Deev Shah. Either way, the male connector and crimps can stay the same

Aarjav Jain (Nov 2025)

@Krish D @Michael Lin: Please check the wire in the ELEC inventory. Ensure twisted pairs are used where needed.

Krish D (Nov 2025)

@Michael Lin I remember you drew out the connections of the 8:2 MUX. Adding this diagram here would make it more clear what you are referring to, since it isn't clear what factors (the connections from the ADBMS1818 GPIOs in particular) are being used to determine the proper input/output ratio <- also the logic for them as well.

Untitled

Author: Krish D

Date: Nov 2025

[@Hemat Wander](#) [@Aarjav Jain](#) [@Michael Lin](#) @Deev Shah

Context

Me and Hemat had an in person ideation session where we thought of a way to route an connector from the slaveboard to the control board for scrutineering access.

The point of this circuitry is to ensure that we can access **all** of our relevant voltage-checking circuitry current circuitry from the control board. This will enable scrutineering process to be more accessible, less complex/procedural , less-HV exposure and more timely (considering we took ~2 hours during FSGP 2025).

Design requirements

Proposed design Concept

Anything in the diagram above that is in white or green represents circuitry that will exist on the slaveboard. Yellow connections literally show the wire.

You may be curious as **when the switches will be toggled**. This can be done by reflashing the system to toggle the switch before the BMASIC polls the voltage values, or by using a physical switch (when pressed will tell the MCU on masterboard to send an ISO-SPI message to toggle the GPIO DURING idle operation) to effectively MUX between the real mod 32 and the external mod 32.

What isn't included in the circuit diagram?

- There isn't any fuses, ESD protection, or by-pass jumpers caps included, since the purpose of the diagram was to make the accessibility circuit more clear. Fuses would be added where the external circuitry connector interacts with the slaveboard (yellow-white nodes).
- There is no connections to show what net/signal will actuate the switches. The signal responsible for doing this would be anyone of the 9 controllable GPIOs on the ADBMS1818.

What implications are there for the rest of the slaveboard?

- The current mux layout (3x 6->3 MUXs for multiplexing temperature inputs) has to be changed since this implementation utilizes the 9 total available GPIOs. This can be mitigated by choosing 1x16->4 MUX which would use 4 GPIOs for controlling the MUX, 4 GPIOs for reading the temperatures, and would leave 1 GPIO available for toggling the switches. This would also change the current firmware we have for toggling specific outputs of the MUX.
- Generally, one other connection + relevant jumpers for bypassing the switches need to be added as redundancy, increasing layout complexity.
- [HV SPDT switches](#) are a tad expensive (~7.40 on Digikey). However only 2 would be required

Failure points of the mechanisms + mitigations

@Hemat Wander @Michael Lin @Aarjav Jain Please let me know your thoughts on complexity, feasibility, however I believe the gains would create a much more elegant & scrutineering process. Do note that currently the slaveboard dimensions are mostly up to us, meaning size of the board can change. A larger board = less routing complexity.

@Deev Shah Note that this may result in an extra cutout, so other methods for routing this external connector the top of the pack in w ay that makes it secured, is most ideal. If this idea gains more popularity and implantation, lets discuss how we can do this in more detail.

Hemat Wander (Dec 2025)

@Krish D

I think that the C17 connection needs a **SPST**, not a **SPDT**. This is because C17 always connects to the 32nd module (+) which we will toggle to connect to the 33rd module (-) outside of the pack.

THEN, for C18, we will need a **SPDT**, which will either connect it to the internal pack 32nd module (+), or to the external 33rd module (+). Note if C18 is connected to the internal 32nd module (+) then BOTH C17 and C18 will be connected to the internal 32nd module (+). This will be normal operation mode as shown in the image in this [reply](#). On the other hand, if we connect C18 to the external 33rd module (+), then we will be in scrutineering mode.

We can now get rid of the thermistor MUX, and just connect one of the pins to a spare GPIO.

Does this make sense/agree with what you were saying? @Krish D

Aarjav Jain (Dec 2025)

@Krish D @Hemat Wander how does this IC work exactly?

[Missing image: ../../images/image_2589921150.png]

Note: We cannot change our FW for scrutineering so we are always reading and faulting on C18. This is why I do not understand why we are using more than 16 cell inputs. That means the scrutineering cell input which is C18 would be 0V which would cause a fault.

So, why are we not lets say switching what we input **into** the slaveboard?

Hemat Wander (Dec 2025)

@Aarjav Jain

That's a good point, I didn't realize we wouldn't be able to change our firmware. Would it be possible to have a scrutineering mode selectable on the master board, via a jumper that we connect for scrutineering mode (monitor C18 voltage) vs not (do not monitor C18).

I say this for two reasons: 1 - Nothing in the scrutineers email mentions if we can / can't do this. 2 - The scrutineers email mentions attaching a spare module to the same measurement **CHAIN** as the rest of the actual modules. Given this, they must mean that we have a spare module connected to a SPARE measurement input. Thus, during normal operation (normal operation = **NOT** scrutineering) that SPARE measurement input couldn't have anything connected to it, simply due to the fact that it is a SPARE. If nothing is connected to it during normal operation, then there is no way that it wouldn't cause a **FAULT unless** we are meant to "de-activate" that input.

Thus, we should be able to de-activate that input.

However: ** If we cannot** deactivate that input, we will need to revert back to the [original scrutineering setup](#).

Krish D (Dec 2025)

@Aarjav Jain I agree with @Hemat Wander idea to add some form of toggling through the masterboard, this would be the most feasible option to activate the switch before the masterboard begins requesting voltage and temp data from the slaveboards.

Toggling idea is great idea @Hemat Wander! Can you draft a reply to Dan? Use the previous email chain to him as inspiration as needed.

Untitled

Author: Hemat Wander

Date: Nov 2025

Notes on Fusing for Slaveboard: As previously discussed in [this update](#), I think that we should add additional fusing to help protect the ASIC as it is connected to high voltage. Previously I mentioned adding additional fuses going to GND from the ASIC to help protect against ESD shorts that cause a high overcurrent internally leading to GND.

However, after discussing with @Michael Lin, we realized that it doesn't make much sense to include this as we have no justification that any overcurrent scenario through the ASIC would go to GND (which is where we would be fusing it). Thus, we will no longer be adding this fuse. The original purpose of adding this fusing was to protect against spikes in current caused from the HV connections to the cell-boards, so we can **instead** take a look at the fusing on that line.

Cell board fuses: Currently we have 750 mA [C1Q 750](#) fuses connected on every module input from the cell-boards, however we found that these fuses didn't stop high current (>5A) ESD scenarios. For reference, we have no ways of knowing if those overcurrent scenarios actually were above 750 mA (all the way to >5A), however [previous LTspice simulations](#) make us believe this.

We discussed increasing the speed of the fuses, however on Digikey it seems like the C1Q750 model is the fastest SMD fuse that is over 600mA. Specifically, it has the smallest I²T value (melting time) for all fuses on Digikey in the 600mA-800mA range. In order to get a faster fuse we would need to get the 500 mA [C1Q 500](#), which is only 4x faster than the current fuse we use. I honestly don't think anything on the ADBMS1818 datasheet gives information about fusing time would be sufficient to protect the chip against over-currents, so I think we should just go with the smallest fuse time that makes sense. As balancing only takes 420mA at max, I think a 500mA fuse would suffice. Thoughts @Krish D @Michael Lin ?

Temperature Circuitry Fuses: The idea behind fusing the temperature circuitry is that on a given cell-board, the cell voltage can be very high relative to the voltage of the temperature circuitry. The temperature circuitry is always 0V-3V above the voltage of the slave boards first module, meaning the voltage of the cell board can be >60V above the temperature circuitry. Thus, in the case we accidentally short something on the cell-board side, especially given that our thermistors will be more involved this time around (through-holes, which are epoxied to the cells). I think there is enough risk where we should include a fuse. To simplify things, the fusing should look like the following, where we only have one fuse for all of the VREF2 and GNDs going to the cell-board.

[Missing image: ../images/image_2566873487.png]

The issue with this, is that if a cell-board shorts we won't know which one it was (slightly longer to find it and fix it), and that we have no fusing on the Tsense line, so if it shorts to the Batt(+) or BATT(-) for some of the other modules, the

voltage will go outside the max voltage range of the MUXes (causing them to break or short, I'm not sure). -

If we did add fusing for each Tsense line it would add 16 additional fuses, which seems like overkill. Thoughts
@Krish D

[Missing image: ../../images/image_2566875626.png]

Other than that, we would have NO additional fusing. All other overcurrents would be protected (hopefully) by the Zener diodes we add between cell inputs.

****Side Note: **** Also I contacted AD about using Zeners and ESD protection circuitry [here](#).

This update took ~30-45 minutes? (I didn't time it)

Krish D (Nov 2025)

@Hemat Wander

A 500mA fuse would suffice. Can you define what the maximum expected current draw is to reference why this is the case?

For reference, we aren't sure yet if the thermistors will be epoxied to the cells or not.

3. Great job on condensing the fuses for the temperature lines. I think this is a much more efficient solution. Given the number of fuses being added (cell board -> SB and Tsense + GND fuse), I thought adding LED indicators for the larger fuses would be a good idea, however this just seems like unnecessary current draw. This being said, debugging the slaveboards will be a more involved procedure since more of the failure modes are being accounted for through our protection circuitry. Consider adding items like this to a "Slaveboard Debugging" guide, since checking continuity across fuses is a bit more implicit.

4. Please remember that the B+ and B- fuses will be on the slaveboard now to increase accessibility.

Untitled

Author: Hemat Wander

Date: Nov 2025

Simulating Auto-Connection Circuitry #2: As mentioned in the [previous update](#), I wanted to confirm that the time delay between circuitry connections was sufficient by testing using the equivalent circuitry going to the ASIC. This includes both the resistors and capacitors leading up the ASIC (seen below) and the ESD protection circuitry within the ASIC (also seen below).

[Missing image: ../../images/image_2564211675.png]

[Missing image: ../../images/image_2564211842.png]

To implement this in LTspice I used the following circuitry. Note that the balancing FET behavior is an approximation that the gate voltage would be pulled up to the next module positive voltage. I should note that this isn't exactly as it would be connected to the ADBMS1818, as the chip has 18 cell inputs, and we only have 16 cells. However the goal of this simulation is just a quick check that the autoconnection circuitry works, which this essentially confirms.

[Missing image: ../../images/image_2564216649.png]

We can now simulate and look at the currents through the ESD protection diodes. We see that the currents stay below 2uA in magnitude, which is within the regular amount we would expect in operation. This shows that the current limit is safe during connection. (For reference, if we connect wires out of order, we can see spikes up to ~8A.

[Missing image: ../../images/image_2564222519.png]

We can also look at the voltages across each capacitor over time, and we see that the voltages stay well under 8V, even during the spike when we connect the switch.

[Missing image: ../../images/image_2564231700.png]

Conclusion The above two pieces of data are justification that the connection circuitry we have chosen is sufficient in that they will not blow up the ASIC (hopefully). I will continue with this circuitry for the schematic!

Next steps for slave board: - Justify fuse timing values from [this update](#). - Send email to Analog Devices about if there is a specification for what Zener diodes we use - Start work on schematic.

The complete LTspice file is attached below (it's pretty messy, my excuse is I did this quickly).

[Auto_connect_experimenting.asc](#)

This Update took 50 minutes.

Krish D (Nov 2025)

Hi @Hemat Wander, great job finding out what the problem was!

Can you make it more clear what exactly what you changed from the previous circuitry and what made it work?

Untitled

Author: Hemat Wander

Date: Nov 2025

Simulating Auto-Connection Circuitry: After performing some simulations for the auto-connection circuitry, I'm more confused than I was before. The goal is that the Gate Source voltage curves for each module should have curves that have steeper slope than the module AFTER it. That way, each PFET would only close after all of the previous PFETs

for the previous modules have already closed.

[Missing image: ../../images/image_2557456578.png]

For the first few modules, we see that the gate source voltage curves look like what we want, meaning that the first module has the steepest curve (green), the second module curve (blue) is in the middle, and the third module curve is the least steep (orange).

[Missing image: ../../images/image_2557458402.png]

However, for the last two modules, for some reason the gate-source voltage curves has a different behavior. The 7th module had a less-steep curve than the 8th module. (The 7th module is in red and the 8th module is in grey).

[Missing image: ../../images/image_2557458546.png]

So in short, the issue is that as we get the higher modules, for some reason the time constant for the capacitors begins to decrease again, although I'm not entirely sure why. To try and learn more about what's happening I decided I should try and focus on the transient conditions as soon as you flick the switch.

We know that the voltage of a capacitor cannot change instantaneously, meaning that the voltage it holds from before the switch gets closed will remain. In this case, the voltage of the PFET is initially pulled up to the cell positive voltage, meaning we can assume each capacitor begins with 4V. We can now simulate with each capacitor now replaced with a 4V source, and then measure the current after the switch gets flipped. We get the following results.

(

Zoomed in photo:

[Missing image: ../../images/image_2563997207.png]

The current through the last two "capacitors" (which are now replaced with voltage sources) can be seen as the following during startup.

[Missing image: ../../images/image_2563998655.png]

The current seems to jump to around 3 mA for the last capacitor while the current through the second last capacitor settles to something much smaller. This shows why the voltage of the last capacitor was dropping so quickly (since the current draw was so high). The reason this current draw is so high is due to the "D8" diode which has a lower on resistance than the 1k capacitor, meaning we get more current flowing from the capacitor, as opposed to from the positive of the V8 voltage source. This tells me that the current flow is so high because of the R16 resistor pulling everything up to V8's positive. So I want to try two things next (both in isolation and together).

1. Try adding a resistor on the diode paths to try and increase the resistance there and even out the current draws.
2. Remove the R16 pullup resistor to the positive voltage of the cells.

For point 1, adding 1k resistors on the discharge line didn't seem to solve the main issue of the discharge curve for some of the higher modules being faster than the previous modules. Such as for module 8 (blue) and 7 (green) down below.

[Missing image: ../../images/image_2564068382.png]

For point 2, if we remove the R16 pulling everything up to the V+, then all of the voltages will fall to GND (0V), which will then exceed some of the PFETs max gate-source source undervoltage thresholds.

I also tried increasing the number of cells to 16 to try and see how a circuit of increased size would behave. Interestingly, the first 8-9 cells seem to behave correctly in that the gate voltage drops fastest for the first cell, and then slightly slower for the second cell and so on. (This is seen with the gate-source voltage curves in the below chart, where the fastest drop occurs for the 1st module, and the second fastest is with the 2nd module, etc.)

[Missing image: ../../images/image_2564114705.png]

Final Product: I last thing I tried was increasing the resistance values of the resistors on the line of the diode for the last 8 modules (modules 8-16), meaning that we would now need 8 additional resistor types (2.2k, 6.8k, 10k, 15k, 22k, 33k, 47k, 66k). However, this shouldn't affect the price, so this is fine - only downside is a little more annoying to bring up, so we can include resistance values on the silkscreen.

The gate-source voltage curves for all of the module PFETs look as follows: Each module has a slower voltage drop time than the previous module, meaning that the circuit seems to work!!

[Missing image: ../../images/image_2564157496.png]

*Note that I'm assuming that the time between each of the PFETs turning on is sufficient for our application. I don't think the datasheet would specify this, however I can check by simulating with the equivalent ESD circuitry of the ASIC (that the current through each diode doesn't spike past the 10mA limit).

Next Steps: 1. As mentioned I want to try simulating with the rest of the ASIC equivalent circuitry to confirm the voltage input won't shock the internal ESD protection diodes. (Confirming that the difference between time constants is sufficient). 2. Although this circuitry seems to work, we might want to compare with options that require fewer components or fewer types of resistors, so I will simulate that next.

(This update took 3 days, 4 hours, and 18 minutes - beating Chris K's record)

| **Aarjav Jain** (Nov 2025)

| @Hemat Wander:

Untitled

Author: Hemat Wander

Date: Nov 2025

Simulating ESD protection circuitry: From the last [update last night](#), I was getting confused why the current through the internal 24V Zener diodes, stayed so high regardless of us using smaller voltage clamping voltage Zener diodes. I found the reason for this is likely due to the 24V Zener diode internal to the LTC having a very small on resistance, compared to the other clamping voltage Zener diodes. (Rs in the picture below).

If we instead use a 33V Zener diode between each 3 inputs, then we will see that the current through the 24V Zener only spikes to around 5A, and most of the current goes through the 33V Zener (at least initially).

The conclusion from this is that the Zener's we choose will only be able to save the LTC based on if they have a smaller on resistance than the ASIC's internal ESD diodes. As this information is not available on the datasheet, we have no way of knowing if this will work.

However, the datasheet recommends using 6.8V zeners between inputs anyways for the grounded capacitor system (we are using differential capacitors), so we might want to consider 6.8V zeners anyways.

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Aarjav Jain (Nov 2025)

@Hemat Wander Good points. Could you email AD and ask about this? If you found that adding the zener is actually doing nothing because the path into the chip is a lower resistance then its **odd** that they still suggest adding a zener. *Correct me if that is wrong. *

So, lets ask them if there is a particular spec that these zeners need to have other than 6.8V Zener Voltage.

Untitled

Author: Hemat Wander

Date: Nov 2025

More slaveboard Circuitry To create the slave board schematic there were a couple of systems we needed to finalize.

Scrutineering Available Cellboard

For one, we need some method of easily being able to complete scrutineering without having to disconnect the 32nd module connector and reconnect a new one. Doing so requires us going into the pack and removing one of the cellboard-slaveboard connectors, which takes a little while. To streamline this process for scrutineering, we should have another method for being able to connect a slave board for testing, without having to disconnect the connections within the pack.

One possible idea is that we have two separate cellboard-slaveboard connectors for module 32, with one actually connecting to the 32nd module, and one going to an extra connector that extends out of the slave board region into the control board area. Then for the part that connects to the ASIC for the voltage tap line and TSENSE lines (circled below), we will have a jumper cap selection system where you can either choose to use the actual 32nd module input, or the input from scrutineering.

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The main problem with this system is that changing the jumper caps will still require taking off the control board to access the modules, which will take a while and slow down the scrutineering process. Thus, it wouldn't really make

sense to include this jumper cap option, as the main time-consuming process would be removing the control board. Once we do that, either switching the module connector from a short harness to a spare long harness, or removing the jumper cap would take around the same amount of time.

With that in mind, I think creating a separate path for connecting a cell-board for scrutineering is only worth it if we can do so while the control-board stays in the pack. Doing so would require a HV connection extending outside of the slave board to an area on the control board where we could choose between either connecting the actual module to the ASIC, or a separate external cell board. To clarify, we would be adding an additional harness with two wires going to the ASIC (for the voltage & temperature circuitry) and with wires going to the 32nd module cell board (for voltage & temperature circuitry). From there, we can now switch between connecting the ASIC to either the actual pack cellboard or some external cellboard for scrutineering. The downside is we are introducing a HV connection to the controlboard that will always be live with 130V, meaning we would have to keep it well-isolated. For now this seems a little risky so I suggest we continue without it, however it is a possible option to consider. [@Krish D](#) Thoughts?

Testing the ESD protection circuitry I ran some tests in SPICE and got the following outputs. Note that this EDS protection circuitry is only for the purpose of clamping voltages outside of the range of the cell inputs. This is because we have directly seen failures before where the INTERNAL ESD protection circuitry for the cell inputs (voltage taps) has failed (when connecting a module out of order), while we have not seen the ESD circuitry for the other pins fail before.

Unfortunately, the simulations showed the my circuitry doesn't seem to work, although I can't quite tell why.

Without the ESD protection circuitry

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With the ESD protection circuitry:

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****Note** I completed these simulations with a 6V Zener diode instead of a TVS diode (since LTspice had no 6V TVS diodes). However from my research, we can assume the behavior to be pretty similar.

As we can see, in both of the cases, the current through the internal Zener diodes of the LTC spikes past 7A during the startup phase, regardless of if we include external ESD protection. My thought process was that including additional external protection diodes with a smaller breakdown voltage would allow them to draw the current instead of the internal Zener, by clamping the voltage, however, this is not the behavior we observe. Instead, we observe that the current through the large 24V zener diodes (D7 & D8) still spikes to around 8A.

I also tried adding smaller voltage Zener diodes in parallel with the 24V zener diodes to see if they would draw some of the current instead, however the 24V zener diodes seem to be taking most of the current no matter what. This is really confusing to me.

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I also tried using TVS diodes and got the same results, (current spiking to 8A in the 24V zener diodes). You can see the results for that below.

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I'm honestly not sure how to move forward with creating the ESD protection circuitry as it seems like there isn't anyway to get it to work in this simulation. We might want to consider putting on external zeners/TVS diodes regardless to serve as additional protection that might work, but if the simulations are set up correctly it seems like the ASIC will experience a current spike either way. I'm not entirely sure what's causing this, any thoughts @Krish D? @Michael Lin ?

We might have to continue without this circuit if we can't figure out how to get it to work correctly? Now that we have auto-connection circuitry, the issue of the voltage taps experiencing a spike in current likely won't be an issue anyways.

Auto Connection Circuitry I also tried simulating the auto-connection circuitry, by using a switch and checking analyzing the voltage drop curves at the gate of each FET. The goal is that the time constant for each FET should be greater than the FET before it, that way they turn on in sequence. However, for some reason we see that the time constant drops for the last two FETs, meaning that they close at the same speed as the first two FETs. I don't really know why this is happening, but I will investigate tomorrow.

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****Side Note: **** - Also as a note for fusing the GND of the ASIC from the [previous update](#), it seems that VREG will draw a max of ~35 mA, which will be the same as the current going through GND (because of continuity of current). So, we can fuse the GND line with a 50 mA fuse, such that if a current spike occurs somewhere shorting internally to ground for an extended period of time we can save the ASIC by blowing the fuse.

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Krish D (Nov 2025)

Hmm. A lot to unpack regarding functionality of TVS diodes. Does AD not have any Spice models for TVS diodes with a 6V rating? Is it worth reaching out to ask for a Spice model?

Doesn't the behavior for the why the ESD protection not working make sense, since the other TVS diodes (the path the high spike should take when clamped) not have a reference? Perhaps probing the voltages between each TVS diode would help make it more obvious if they are floating or not. Since they are floating, it makes sense why they wouldn't be functioning.

It would make more sense to connect all the modules in order (in the simulation itself) and then try to vary the input signal to a high voltage very quickly (to simulate noise), and then look at the voltage/current behavior across the internal zener diodes and across each TVS diode. This will protect against transient events that occur when the board is powered on properly, however it will not act as a failsafe against protecting modules out of

order UNLESS they are connected to gnd instead of the previous module. This would mean that every TVS diodes clamp rating would change, but we can talk about this more and analyze how they should change.

Does this make sense @Hemat Wander ?

Untitled

Author: Hemat Wander

Date: Nov 2025

Determining necessary protection circuitry for Rev 2.0 The purpose of this update is to discuss some of the research I will do into determining how we will protect the next slave board from small overcurrent events and ESD.

Purpose: The previous slave board suffered from issues with the LTC6813 consistently breaking due to a spike in the voltage or current, meaning we would have to replace it. For example, you connect a random module jumper without connect the previous modules in order, it will create a spike of voltage at that input (say C5), thereby causing a spike in current due to the internal ESD protection in the IC -> which causes an overcurrent that breaks the IC. Furthermore, we've also experienced behavior with the IC something just not working even if we connect everything properly -> thus we suspect that ESD spikes from us working with the PCB and chip caused it to break -> we want to prevent this for the next pack.

Furthermore, we also want to add some fuse protection to protect the ASIC we use from any overcurrent situations we might experience. Both in general from connecting modules, but also in terms of adding protection on the temperature line.

****Goals: ****

1. We want to determine what type of protection diodes we want to use (TVS, ESD, part number, etc.) and where they should be placed.
2. For fusing, we want to determine where the current will go in an overcurrent case, and also what type of fuses we should use (fast acting?, part number?)

Protection Diode Research: The datasheet says that the maximum limit for adjacent Cell inputs is positive 8V, and -0.3V. Ideally, we should include protection against both of these limits, to protect against either a module being connected out of order (ignoring our auto-connection circuitry), and to protect against a module being plugged in reverse. However, it doesn't really make sense to account for the latter with a diode as we would essentially be shorting the module positive and negative if it was connected backwards (not a good idea).

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With this in mind, we should use some sort of voltage protection in the range of 5V-7V between each module, the idea being that if a module has a spike in voltage, the current will flow down to the other modules instead of flowing through the ASIC and breaking it. Also, the only application note related to using diode protection in the ADBMS1818 datasheet is this. It makes sense to include the diode protection like this, to protect against overvoltage's between

modules, so we will do that as well.

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What type of diode should we use (TVS diodes or ESD diodes)? From a quick search it seems that TVS diodes are predominantly for power supply surges which can be for a higher amount of power for a (slightly) extended period of time, while ESD diodes are for protecting against a smaller electrostatic discharge but with a faster time. For our purposes, I think that we want to worry about accounting for a higher amount of power for an extended time, caused by module voltages being connected in the wrong order (TVS diodes). This should also provide some ESD protection as well though. Furthermore Zener diodes are used to consistently regulate a voltage at some higher level, meaning they don't really apply to our use case as our input should usually be lower than the diode breakdown voltage.

After looking on Digi-key, it seemed like there weren't many results with a 5V breakdown voltage, however I found some options with a 6V breakdown voltage. Specifically the [ESD7481MUT5G](#), which we will connect between each of the cell inputs. However, I want to confirm that these diodes will remain as an open circuit for voltages below 4.2V.

Note that we still need to find some way of protecting against modules being plugged in reverse (however we can solve that issue with other methods).

Additional Fuse Protection: On top of the cell-tapping fuses we already have on the cell board, I think that it would be good to add additional fuses to

1. Protect the ASIC from extended current spikes that would stress the diode protection
2. Protect the temperature lines from overcurrent caused by shorting with the module voltages.

For point 1, I don't think we need to do anything overkill, this simply accounts for the difference in the maximum current we can draw through the ASIC (10mA according to the above table), and the fuse rating of the cell board fuses (> 400 mA). We could add one fast acting fuse on the GND connection of the ASIC, with a current rating slightly above that of the max expected current draw from all of the chip systems. -> I still need to determine how to calculate this.

For protecting the temperature lines, we should consider the possibility of the module voltage shorting through the temperature line based on however we choose to implement the thermistor. The circuitry seems like it will be more involved than last time, so I think the severity x likelihood of the risk is large enough to consider adding fusing. The shorts can either occur on the module voltage to the 3V ref, or from the module voltage to GND, or directly shorting the 3V ref to GND somewhere in the temperature circuitry. If an overcurrent occurred, the current draw would be very high, meaning it might suffice to have one fuse on both the GND and VREF (3V) for all of the thermistor circuitries combined (2 fuses total).

The main benefit to have multiple fuses is that we can use the fuses to determine at which module exactly the short occurred. Because the fuses are relatively expensive (~>\$1), but it would be very helpful to know which module shorted, this is a bit of a tough problem.

My conclusion is that we should fuse once on the VREF side and once on the GND side of the thermistor circuit as it would be cheaper, while still achieving the larger goal of having fuse protection for extreme short cases. Each of these fuses will be 28mA-50mA, as the expected current draw from the VREF line is only $16 \cdot (3V/20000 \text{ ohms}) = 2.4 \text{ mA}$. They will be fast-acting SMD fuses.

@Hemat Wander

Hemat Wander (Nov 2025)

To add some more detail and answer your questions:

For the fusing, we know that the ADBMS pins for the temperature circuitry (VREF2, GPIO1-9 if we use them) have an absolute limit at 10mA, meaning that our rating for the fusing has to be below that to protect the ADBMS. However the issue is that when looking on Digi-key for SMD fuses, they are all rated at 28mA or over.

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If looking at non-SMD fuses with the ratings we would want, they seem to be way too-expensive. (\$44).

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It seems like there is no feasible way to 100% protect the ADBMS in the case that high voltage shorts to Tense or VREF2. However, in the case that high voltage does somehow short to one of the temperature sensing lines, we still need fusing, as even if it cannot protect the ASIC it will stop an extended overcurrent situation caused by the short, which would otherwise draw an insane amount of current from the cells and light the cells on fire. (Ex. Module 23(+) = ~92V gets shorted to VREF = ~3V).

With this in mind, we know that the maximum current the ADBMS will be able to draw from each pin is 10 mA, meaning the expected current draw by definition has to be less than 10mA. From there, Digi-key has a few options, but in general we can assume that whatever shorts happen would be over 1 A (if a short happens). Regardless, if we want to use the smallest available fuse that is still above the expected current draw, we can just use a 28 mA fuse, like here

To answer your questions:

1. The current flowing down to the other modules is intended. What this means is that whenever we connect a higher voltage module without connecting the previous ones, the TVS diode will act as a path to first fill up the capacitors of the previous modules with voltages decrementing in 6V between modules down until it reaches 0V, after which all of the modules before that would have 0V. This essentially protects the IC from the 8V limit between cells by filling up the previous module capacitors to the correct voltage. Once the capacitors are filled, current will stop flowing. Note that this protection is for voltage threshold between cell inputs, which is different from ESD caused by static electricity.

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2. The case of a higher amount of power is what I explained before, with the voltage protection for the cell inputs from the previous question using Zener's. Essentially, if we connect modules with the incorrect voltage, the ASIC will experience an extended period of overvoltage between cell inputs -> which we should account for.
3. It is not the only solution, we can instead use two fuses (one on VREF2 and one on GND), which will only cost us ~\$3, as opposed to fusing on each line. That just means we won't know at which module the short happened exactly.

4. Let me know if this explanation doesn't suffice and I will draw it out: If we short one of the higher voltages to the GND of the temperature sensing circuitry on the cell-board, then we will have a lot of current flowing through the GND wires of the slaveboard-cellboard harnesses, which then feed into the GND on the slaveboard, which connects to the first module of the respective slaveboard. If we fuse this line, we can stop the current draw on that line.

Untitled

Author: Hemat Wander

Date: Nov 2025

Voltage Tapping Auto-Connection Circuitry Update #3:

Going on from the previous [update](#):

It's beginning to seem like the only method we have for automatically connecting the voltage taps in order is the PFET circuitry given at the end of the [first update](#). However, before we confirm this I wanted to do some more thinking on possible alternatives, or getting the method from the previous update to work.

For context of the previous method, it currently looks something like this. The idea is that the voltage from the previous cell should feed into controlling the very next cells FET (which connects the cell to the LTC). The issue with this design is that the voltage from the next cells in the series feed backwards into the previous cell and so on forming a voltage divider ladder down to ground voltage. The issue with this is that plugging in any cell later in the chain (say cell 4 with 16V) will cause the voltage of the previous cells inputs to immediately become some fraction of that, despite that cell being missing (cell 3 will read 12V despite the voltage tap not being connected to the actual cell).

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How can we fix this? We tried thinking of ways of integrating diodes into this circuit, but the issue is that this doesn't really work. We need to somehow pass a lower voltage from the left side to the right side without passing a higher voltage from the right side to the left side. A diode works by passing a higher voltage to the other side or in other words passing a lower voltage through backwards.

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I also considered passing the voltage from the LTC input of the previous module, instead of the cell voltage directly. The idea being that the PMOS wouldn't have the voltage at the source of a given module passed back through the previous modules source affecting the previous modules. Also, this makes more sense in terms of working like an "AND" gate, where the voltage only passes through if all the previous voltages are

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This concept seemed promising as I thought the PMOS would act as an open circuit when the connection is floating, thus not allowing the cell voltage to pass through, or propagate backwards. However, after simulating in LTspice, it

became clear to me that this wouldn't be a viable option due to the body diode of the PFET. the body diode of the PFET allows for current to flow backwards from drain to source when the drain voltage is higher than the source voltage. This is essentially always true in our setup, due to the next cell connection increasing the voltage of the previous LTC input.

For example, Cell 2(+) increases the voltage of C1 in comparison to Cell 1(+), thus allowing current to flow through the PFET in reverse bias.

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When simulating, I got that C1 was essentially always slightly higher than 4V despite the input to cell 1 (+) being floating (as opposed to being. Note that the drain and source are flipped in the SPICE model.

After simulations and thinking about more configurations, it doesn't seem like there's any promising way to get this circuitry to work the way we want.

As a reminder our goals are:

- Only connect each PFET when all of the previous modules are connected

- To not propagate voltage of the each modules backwards to fill in the previous module voltage inputs.

The second goal is important for completing open wire checking (as otherwise the previous modules will make it seem like a cell is connected even though it is not).

HOWEVER, I should mention that all of these circuitries should in theory NOT fry the ASIC, thanks to the voltage resistor ladder present in some form or another.

It seems like the conclusion from all of this is to continue with circuitry similar to FEs. The next steps are to discuss this in the slave board check in meeting (I will set this for Tuesday (tomorrow)).

Untitled

Author: Hemat Wander

Date: Nov 2025

Going on from the previous [Update](#):

After doing some more research, I think there are a few other promising methods for being able to connect the voltage taps automatically in order.

One method would be to use this timer block debounce IC (LTC6994), that would allow us to specify an amount of delay time between each of the gates activating. The idea with using this IC is that would have it separately connected to each MOSFET, and then drive the signal to the gate of each MOSFET using one switch with each IC

having different delays.

<https://www.analog.com/en/products/ltc6994-1.html>The ONLY problem I can see with this solution, would be that each IC is so expensive, meaning it would cost ~\$160 per slave board (that's not gonna work for us).

Another method I was thinking of was using some sort of OPAMP circuitry to connect the voltage taps, such that we wouldn't require any FETs. The idea would be that we only power the rails of the MOSFET (turn it on) using one GND connection for each MOSFET, to act as another sort of switch.

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However,

had an idea that we can take the finalized RC circuitry from the previous update, and remove the need for a switch. The idea is that each module connection requires a voltage from the previous cell in order to activate the PFET. This would be a nice solution, the only problem is that it doesn't work.

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The issue, is that we essentially form a voltage divider going from the higher cell voltages to ground, meaning that each cell would get automatically connected.

Krish D (Nov 2025)

@Hemat Wander @Michael Lin Some notes & questions on your ideas:

Regarding your implementation idea for the [LTC6994](#), I think there is much more complexity involved since you are coupling together a MOSFET and this IC (essentially doubling the number of ICs needed to drive the gates). I'm unsure of how you thought to connect with these ICs but I'd also assume there is issues with high voltage isolation since the IC will be reference ground (0V or 60V - depending on the slaveboard you are connected to) and the gate of the FET has a risk of being connected to much larger voltages.

I'm not too sure what the issue is that you are describing. A different issue I see however is that if a module is connected out of order, it will activate the FETs at multiple other points, which may be unwanted. Could diodes not prevent the current from flowing unwantedly? Additionally, you are losing the ability to toggle on and off the connection since the connection is now dependent on if a connector is in or not. Do we need the connection to be toggleable (with a jumper for example)?

Untitled

Author: Hemat Wander

Date: Nov 2025

****Determining auto-connection circuitry: ****

The Problem: With the LTC6813, it was necessary to plug the voltage tap connections in order. The voltage taps refer to the "cell inputs" in the datasheet, which are used to connect the module positives to the LTC and measure the voltage across each module. "Connecting in order", meant that we had to connect each of these voltage taps beginning at the lowest module and going up (module #1, then #2, etc. -> with voltages ~4V, ~8V, ~12V, etc..)

To achieve this on the previous slave board iteration, we had to use the module jumpers (module 13's is shown), such that every time we would connect the slaveboard in the pack we would connect the jumper cap going on each of the module jumpers in order. The process was slow and prone to human error -> thus we want some way to automate this process.

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The Solution

The idea in general should be something similar to what formula electric does, such that we have some sort of timing system to connect the voltage taps in order. Using some sort of RC timer circuitry should fit this purpose. I couldn't find any sort of mechanical solution that could do this for us (@Krish any thoughts?), but I think using an electrical solution would be less prone to error -> similar to what formula E uses.

We need two things to achieve this:

- An electronically controlled switch

- Some timing / ordering circuitry for when to control those switches.

Electronic Switch:

Currently a MOSFET seems like the best option, my only concern is that a FET is not exactly like a switch when its on. Specifically, it might behave weirdly when we try to push current in the opposite direction, such as for open wire checking, or depending on how the ADC reading operates.

This is the more complicated part, as we need to find some way of activating the MOSFETs in order for all of the different voltage inputs, just using one switch. From FE, we figured out we can do this using RC "timers" where we use a certain resistance and capacitance to determine how long the gate voltage takes to reach the V_{gs} threshold.

- Note: I considered having the RC timers connect to comparators that trigger a digital signal (high or low) to the MOSFET to have a specific digital cutoff time, as opposed to a continuous analog increase, however I think as long as the time constant calculations are done correctly, the analog solution will be much cleaner than adding comparators.

My first guess was to use NMOSs and have their gates connected through a resistor ladder to the max voltage, however this would invert all of the time constants (meaning we would have the maximum time constant for the first module, and the minimum time constant for the last module). -> This solution doesn't work

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One simple solution would be to have each FET wired separately to the switch with a varying resistance voltage divider, or a varying capacitance to change the output current. Two possible configurations of this kind are shown

below. In the first one we have a ladder of resistors connecting to the voltage divider circuitry for each. One issue with both of these is that they require a large variety of different resistor values, and they seem a little bit more messy.

Also, note that with the resistance values written they would have quiescent current draws of approximately.

$(4V)/(16000 * 3 + 5000 \text{ ohms}) = 6.35 * 10^{-5} \text{ A}$ between each module

and

$(64V)/(16000 * 3) = 1.3 \text{ mA}$ between module 1 and 16

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To try and make a cleaner solution, I used PFETs to invert the logic of the first solution. This time we switch the GND connection as opposed to the Vmax connection. However, the issue with this circuitry, is that in the off state (switch disconnected) the 100ks will have a much higher resistance than the 1k ohm ladder chain, meaning that the voltage at each gate will be heavily influenced by the other module voltages.

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To fix this final issue, we can add diodes at the gate of the MOSFETs essentially like how Formula E does there's. These diodes stop the current from flowing from the other modules to the gate of each given module.

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We basically ended up with what Formula E has, but I wanted to derive what they had from first principles, as opposed to blindly copying it.

Next Steps: Before moving forward implementing in the schematic we might consider: 1) Making a spice simulation to ensure it works the way I think 2) Trying to consider other circuit topologies. Essentially just continue brainstorming concepts like I did here to see if any other solution is cleaner.

Krish D (Nov 2025)

Great job on breaking down your thought process. I appreciate that you took the time to break this down since I was considering these ideas as well.

I think doing any form of a mechanical solution would be quite space intensive and more prone to failure, so using semiconductors as switches would likely be the best bet for a robust solution.

I believe both of your solutions (images 6,-7) are feasible to do. You mentioned that there would be a larger quiescent current draw, however if diodes are put in place, this could be avoided. While diodes would make your two solutions work, there is still many more resistors associated with them, so ultimately going with the last solution seems like a better idea to me.

@Hemat Wander @Michael Lin Did you folks manage to come up with another solution yesterday? If so, please document it here when you can, but otherwise this justification for choosing the circuitry makes sense.

Aarjav Jain (Nov 2025)

"We basically ended up with what Formula E has, but I wanted to derive what they had from first principles, as opposed to blindly copying it." -> Haha. This is beautiful though and exactly how our engineering work should be done. So, as Krish said, this is great that you derived it. Every circuit and component **needs** a justification at Solar.
